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Locality of turbulent cascades

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Abstract

In this paper we establish sufficient conditions for locality of turbulent cascades, by an exact analysis of the fluid equations. No assumptions of homogeneity or isotropy are invoked and the results apply to individual realizations with no resort to any statistical averaging. The only requirement is suitable regularity of the turbulent solutions of the fluid equations in the high Reynolds number limit, corresponding to Hölder continuity but non-differentiability in space. We use a smooth filtering approach to resolve the turbulent fields both in space and in scale. We discuss several physical cascades to exemplify and clarify the analysis, including joint energy and helicity cascades in three space dimensions and the dual inverse energy cascade and direct enstrophy cascade in two dimensions.
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1. Introduction

One of the key ideas in turbulence theory is that of *locality* of cascades. The essential idea is that only modes near a given scale contribute to transfer across that scale. The concept has a central importance in the subject because locality is one of the conditions invoked to justify universal statistics at small scales. If excitations are transferred by a chain of chaotic steps from scale to scale, then the conditions at the large scales may be forgotten and only the interactions at a sequence of adjacent scales will determine the characteristics at small scales.

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In this paper we shall establish sufficient conditions for scale locality of turbulent cascades, in terms of regularity properties of the solutions of the fluid equations as viscosity tends to zero. Among the examples that we shall consider are energy and helicity cascades for incompressible fluid turbulence in three dimensions, and energy and enstrophy cascades in two dimensions. The regularity that we shall assume for Euler solutions in the high Reynolds number limit is the Hölder type that was conjectured theoretically [1,2] and is observed (in a space-mean sense) experimentally [3]. Thus, we shall show that cascades are scale-local when pointwise Hölder exponents lie between certain bounds, or, alternatively, when structure-function scaling exponents lie between certain bounds. More precisely, we shall show that cascades are local when the fluid solutions have space-regularity in certain *Besov spaces* [4] that consist of continuous but non-differentiable functions. The results presented here are an improvement of those that we established earlier using Fourier methods [5] and are based instead upon the filtering approach [6] to scale-resolution in turbulence. We used this approach before to establish estimates on spatially-local fluxes [7] and the present work shall make essential use of that previous analysis.

The physical intuition behind locality is very simple. For example, the flux of energy across a length-scale ℓ can be represented by a product of strain $\bar{S}_{ij}$ from large scales $>\ell$ and of stress τ_{ij} from small scales $<\ell$:

$$\Pi(\mathbf{x}, t) = -\bar{S}_{ij}(\mathbf{x}, t)\tau_{ij}(\mathbf{x}, t).$$

This quantity represents the rate of work done by the large scales against the small scales. Precise definitions will be given below. The only essential point here is that the strain due to velocity modes at length-scale $r > \ell$ is of the order $S(r) \sim \delta u(r)/r$ and the stress due to modes at length-scale $r < \ell$ is of the order of $\tau(r) \sim (\delta u(r))^2$. The quantity $\delta u(r)$ is the magnitude of the velocity increment across two points separated by a distance r. Now suppose that the velocity increments scale as $\delta u(r) \sim r^\alpha$ for some Hölder exponent $0 < \alpha < 1$. Then it follows that

$$\frac{S(r)}{S(\ell)} \sim \frac{r^{\alpha-1}}{\ell^{\alpha-1}} \sim \left(\frac{\ell}{r}\right)^{1-\alpha}, \quad r > \ell. \tag{1}$$

Thus, most of the strain comes from $r \approx \ell$, when $\alpha < 1$. We call this property *infrared locality*. Likewise,

$$\frac{\tau(r)}{\tau(\ell)} \sim \frac{r^{2\alpha}}{\ell^{2\alpha}} \sim \left(\frac{r}{\ell}\right)^{2\alpha}, \quad r < \ell. \tag{2}$$

Therefore, most of the stress comes from $r \approx \ell$, when $\alpha > 0$. This property we call *ultraviolet locality*. In short, the cascade is local in scale and only involves modes near the length-scale ℓ when the Hölder exponent satisfies the bounds $0 < \alpha < 1$.

In the discussion below we shall make rigorous the above argument and, as well, generalize it to a wide variety of other turbulent cascades. The organization of the paper is as follows. In Section 2 we introduce the filtering approach to simultaneous space-scale resolution for turbulent fluids. We shall discuss several cases of turbulent cascades by means of this technique, which will be used as examples later in the paper. The main results of the paper are presented in Section 3, where we formulate and prove the rigorous locality estimates. For the sake of clarity, we first carry out the proofs under the simplified assumption of monofractality and establish locality in terms of pointwise Hölder exponents. Next, we extend the analysis to general multifractal velocities, with conditions on Besov space exponents. In Section 4 we discuss several concrete cases of physical interest: energy and helicity cascades in three space dimensions (3D) and energy and enstrophy cascades in two dimensions (2D). These examples not only illustrate the general results, but also help to clarify some finer points. In particular, the enstrophy cascade in 2D will be shown to be, in fact, infrared nonlocal. In Section 5 we shall discuss the relationship between our results and those of previous authors. Finally, in Section 6 we mention some important directions for future work.

2. The filtering approach and local flux

Following Germano [6], we resolve turbulent fields simultaneously in space and in scale using a simple filtering approach. Thus, we define a smooth low-pass filter

$$\bar{\mathbf{u}}(\mathbf{x}) = \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\mathbf{u}(\mathbf{x}+\mathbf{r}), \tag{3}$$

where G is a smooth mollifier or filtering function, nonnegative, spatially well-localized, with unit integral

$$\int \mathrm{d}\mathbf{r}\, G(\mathbf{r}) = 1. \tag{4}$$

The function G_ℓ is rescaled with ℓ, as

$$G_\ell(\mathbf{r}) = \ell^{-d} G\left(\frac{\mathbf{r}}{\ell}\right), \tag{5}$$

where d is the space dimension. Thus, (4) holds also for the rescaled function and the modes at length-scales $<\ell$ have been averaged out in (3). Likewise, we can define a complementary high-pass filter by

$$\mathbf{u}'(\mathbf{x}) = \mathbf{u}(\mathbf{x}) - \bar{\mathbf{u}}(\mathbf{x}). \tag{6}$$

In the case of wall-bounded turbulence, one may wish to use a filtering function which depends upon the point $\mathbf{x}$ considered, as $G(\mathbf{r};\mathbf{x})$ and the convolution in (3) is replaced by a more general integral transform. This generalization can be handled, but at the price of additional complications, such as the non-commutation of differentiation and filtering operations [8]. To avoid these, we consider turbulence only in the bulk of the domain at a fixed (but arbitrary) distance from the wall.

If the above filtering operation is applied to the incompressible Navier–Stokes equation

$$\partial_t \mathbf{u} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla p + \nu \triangle \mathbf{u} \tag{7}$$

with $\nabla\cdot\mathbf{u} = 0$ determining the pressure p, then one obtains

$$\partial_t \bar{\mathbf{u}} + (\bar{\mathbf{u}}\cdot\nabla)\bar{\mathbf{u}} = -\nabla\cdot\boldsymbol{\tau} - \nabla\bar{p} + \nu\triangle\bar{\mathbf{u}}, \tag{8}$$

where

$$\boldsymbol{\tau} = \overline{\mathbf{u}\mathbf{u}} - \bar{\mathbf{u}}\,\bar{\mathbf{u}} \tag{9}$$

is the *stress tensor* from the scales $<\ell$ removed by the filtering. It is actually the tensor $\varrho\boldsymbol{\tau}$, multiplied by mass density ϱ, which is a stress (spatial momentum flux) but we take the usual liberty with the use of the term. The Eq. (8) is valid even if the Navier–Stokes equation has singular solutions, interpreted in the sense of distributions. In that case, (8) holds with classical derivatives in space and distributionally in time. In fact, this is equivalent to the standard notion of "weak solution" of a partial-differential equation with solutions too singular for derivatives to exist in the classical sense (e.g. see [9], Section 11.1.1). Similar remarks apply to singular or distributional solutions of the incompressible Euler equations obtained in the inviscid limit, in which case the terms proportional to ν are absent.

The equation for energy balance in the large scales is [7,10]:

$$\partial_t e + \nabla\cdot\mathbf{J}_E = -\Pi_E - \nu|\nabla\bar{\mathbf{u}}|^2, \tag{10}$$

with large-scale *energy density*

$$e = \tfrac{1}{2}\bar{\mathbf{u}}^2, \tag{11}$$

spatial *energy transport* vector in the large scales

$$\mathbf{J}_E = (e + \bar{p})\bar{\mathbf{u}} + \bar{\mathbf{u}} \cdot \boldsymbol{\tau} - \nu \nabla e, \tag{12}$$

and scale-to-scale *energy flux*

$$\Pi_E = -\nabla \bar{\mathbf{u}} : \boldsymbol{\tau}. \tag{13}$$

As above, this equation should be multiplied through by density ϱ to give the true energy balance, and for Euler solutions the terms proportional to ν should be dropped. It is important to notice that there is an arbitrariness in the definition of the "sink" term on the righthand side of Eq. (10). For example, we could define instead

$$\mathbf{J}'_E = (e + \bar{p})\bar{\mathbf{u}} - \nu \nabla e, \tag{14}$$

and

$$\Pi'_E = -\bar{\mathbf{u}} \cdot (\nabla \cdot \boldsymbol{\tau}). \tag{15}$$

In fact, there are infinitely many possible ways to shift terms between the divergence term on the left and the sink term on the right. This is a well-known ambiguity in the definition of "dissipation functions" in nonequilibrium thermodynamics ([11], Chap. III, Section 2). Here it is crucial that we adopt the original definitions in (12) and (13), rather than those in (14) and (15), or others. In fact, we shall see later that while the energy flux defined in (13) – the unique Galilei-invariant choice – is scale-local, that in (15) is not. The same lack of scale-locality shows up with the original definitions, but there as a property of the space transport vector (12). This makes sense physically, because nonlocal convective sweeping does change the local energy density, but only by moving energy from point to point in space. It is crucial that the definitions (12) and (13) allow all such non-local interactions to be subsumed into space-transport. In particular, all such contributions cancel in a space integration of (10) over the volume of the flow, with boundary conditions that correspond to zero flow of energy through the wall. Thus, such non-local interactions are present but give no net transfer of excitation to the small scales $<\ell$.

Similar results hold also for helicity balance in the large-scales. If one takes the curl of (8) to obtain an equation for the large-scale vorticity vector $\bar{\boldsymbol{\omega}}$,

$$\partial_t \bar{\boldsymbol{\omega}} + (\bar{\mathbf{u}} \cdot \nabla)\bar{\boldsymbol{\omega}} = (\bar{\boldsymbol{\omega}} \cdot \nabla)\bar{\mathbf{u}} - \nabla \times (\nabla \cdot \boldsymbol{\tau}) + \nu \triangle \bar{\boldsymbol{\omega}}, \tag{16}$$

then it is not hard to derive [12]:

$$\partial_t h + \nabla \cdot \mathbf{J}_H = -\Pi_H - 2\nu \nabla \bar{\mathbf{u}} : \nabla \bar{\boldsymbol{\omega}}, \tag{17}$$

with large-scale *helicity density*

$$h = \bar{\mathbf{u}} \cdot \bar{\boldsymbol{\omega}}, \tag{18}$$

spatial *helicity transport* vector in the large scales

$$\mathbf{J}_H = h\bar{\mathbf{u}} + (\bar{p} - e)\bar{\boldsymbol{\omega}} + 2\bar{\boldsymbol{\omega}} \cdot \boldsymbol{\tau} - \bar{\mathbf{u}} \times (\nabla \cdot \boldsymbol{\tau}) - \nu \nabla h, \tag{19}$$

and scale-to-scale *helicity flux*

$$\Pi_H = -2\nabla \bar{\boldsymbol{\omega}} : \boldsymbol{\tau}. \tag{20}$$

Again the definitions in (19) and (20) are well-chosen so that the flux in (20) is scale-local, as shown later, and all of the nonlocality is isolated in the space transport vector (19).

Another example is provided by the cascades of energy and enstrophy in 2D. The energy balance does not differ in form from that in (17) for 3D. However, the enstrophy balance is more easily derived from the 2D incompressible Navier–Stokes equation written in terms of the (pseudoscalar) vorticity $\omega = \mathbf{e}_3 \cdot (\nabla \times \mathbf{u})$:

$$\partial_t \omega + (\mathbf{u} \cdot \nabla)\omega = \nu \triangle \omega. \tag{21}$$

Filtering this equation gives

$$\partial_t \bar{\omega} + (\bar{\mathbf{u}} \cdot \nabla)\bar{\omega} + \nabla \cdot \boldsymbol{\sigma} = \nu \triangle \bar{\omega}, \tag{22}$$

where

$$\boldsymbol{\sigma} = \overline{\omega \mathbf{u}} - \bar{\omega}\,\bar{\mathbf{u}} \tag{23}$$

represents the space transport of vorticity due to the eliminated small scales $<\ell$. It is then very straightforward to derive the large-scale balance of enstrophy [13]:

$$\partial_t \zeta + \nabla \cdot \mathbf{J}_Z = -\Pi_Z - \nu |\nabla \bar{\omega}|^2, \tag{24}$$

with large-scale *enstrophy density*

$$\zeta = \tfrac{1}{2}\bar{\omega}^2, \tag{25}$$

spatial *enstrophy transport* vector in the large scales

$$\mathbf{J}_Z = \zeta \bar{\mathbf{u}} + \bar{\omega}\boldsymbol{\sigma} - \nu \nabla \zeta, \tag{26}$$

and scale-to-scale *enstrophy flux*

$$\Pi_Z = -\nabla \bar{\omega} \cdot \boldsymbol{\sigma}. \tag{27}$$

In this case we shall see that there is essential non-locality in the transport vector (26) but also in the flux (27) and thus no shifting of terms between them can eliminate the nonlocality.

All of the examples that we have considered can be cast into a more general framework, by defining the "flux"

$$\Pi(\mathbf{u}, \mathbf{v}, \mathbf{w}) = -\bar{\nabla}\mathbf{u} : \boldsymbol{\tau}(\mathbf{v}, \mathbf{w}), \tag{28}$$

with the notations

$$\bar{\nabla}\mathbf{u} = \nabla \bar{\mathbf{u}}, \quad \boldsymbol{\tau}(\mathbf{v}, \mathbf{w}) = \overline{\mathbf{v}\mathbf{w}} - \bar{\mathbf{v}}\,\bar{\mathbf{w}}. \tag{29}$$

In particular, the enstrophy flux in the third example fits also into this framework if we define the vorticity vector to be $\boldsymbol{\omega} = \omega \mathbf{e}_3$ there. Thus,

$$\Pi_E = \Pi(\mathbf{u}, \mathbf{u}, \mathbf{u}), \quad \Pi_H = 2\Pi(\boldsymbol{\omega}, \mathbf{u}, \mathbf{u}), \quad \Pi_Z = \Pi(\boldsymbol{\omega}, \boldsymbol{\omega}, \mathbf{u}). \tag{30}$$

In fact, the tensorial character of the quantities $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in (28) is not so important and we shall assume them to be vectors just for convenience. Notice that $\boldsymbol{\tau}(\mathbf{v}, \mathbf{w})$ defined in (29) is bilinear and symmetric in the sense that $[\boldsymbol{\tau}(\mathbf{v}, \mathbf{w})]^\top = \boldsymbol{\tau}(\mathbf{w}, \mathbf{v})$. Furthermore, $\Pi(\mathbf{u}, \mathbf{v}, \mathbf{w})$ is trilinear in its three arguments, because the large-scale gradient $\bar{\nabla}\mathbf{u}$ is a linear functional. In the next section we shall show that each of the quantities $\bar{\nabla}\mathbf{u}$, $\boldsymbol{\tau}(\mathbf{v}, \mathbf{w})$, and $\Pi(\mathbf{u}, \mathbf{v}, \mathbf{w})$ is scale-local, in a sense there made precise, assuming suitable space-regularity of the fields $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$.

3. Proof of scale locality

We here prove locality for the general quantities above, using conditions on the regularity exponents of the fields involved, in the limit as viscosity tends to zero.

3.1. Monofractal fields

Let us first assume that the fields $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are *monofractal* with Hölder exponents α, β, γ, respectively. More precisely, for each space-time point $(\mathbf{x}, t)$ we define the *pointwise Hölder exponent*

$$\alpha(\mathbf{x}, t) = \liminf_{\mathbf{r}\to\mathbf{O}} \frac{\log|\delta\mathbf{u}(\mathbf{x}, t; \mathbf{r})|}{\log|\mathbf{r}|}, \tag{31}$$

where

$$\delta\mathbf{u}(\mathbf{x}, t; \mathbf{r}) = \mathbf{u}(\mathbf{x}+\mathbf{r}, t) - \mathbf{u}(\mathbf{x}, t) \tag{32}$$

is the increment of $\mathbf{u}$ at $(\mathbf{x}, t)$ for the space vector separation $\mathbf{r}$. This is the standard definition used in the rigorous mathematical theory of multifractal functions. See [4,14]. Notice that this quantity corresponds to the supremum of the values of α for which $\mathbf{u}$ belongs to the local Hölder class $C^{\alpha}(\mathbf{x}, t)$. Of course, if we take the limit infimum in the definition (31) at a fixed positive viscosity $\nu > 0$, then presumably $\alpha(\mathbf{x}, t) \equiv 1$ everywhere.[1] Instead, we are considering the exponents that are obtained from (31) after first taking $\nu \to 0$ and then $\ell \to 0$. In that case, it is expected [2] that one obtains instead a nontrivial function $\alpha(\mathbf{x}, t)$ with values in an interval $[\alpha_{\min}, \alpha_{\max}] \subseteq [0, 1]$. A variety of methods have been proposed to extract these exponents from experimental data at finite Reynolds number when there is scaling for only a finite range of length-scales, e.g. see [15].

To make things simple in our discussion below, we shall assume that there is a single value α such that $\alpha(\mathbf{x}, t) = \alpha$ for all $(\mathbf{x}, t)$, and similarly for exponents β, γ of $\mathbf{v}, \mathbf{w}$. Fields which satisfy this assumption are called *monofractal*. It is believed, for example, that the velocity field in the 2D inverse energy cascade is monofractal with exponent $\alpha = 1/3$. Using the simplifying assumption of monofractality, we establish locality, first for $\bar{\nabla}\mathbf{u}$ and then for $\boldsymbol{\tau}(\mathbf{v}, \mathbf{w})$, with suitable conditions on the Hölder exponents of $\mathbf{u}, \mathbf{v}, \mathbf{w}$.

3.1.1. Locality of the large-scale gradient

In [7] a simple estimate was established for $\bar{\nabla}\mathbf{u}$ in terms of the pointwise Hölder exponent. We must first remind the reader of the proof of that result. It is straightforward to show (dropping the time variable for simplicity) that

$$\bar{\nabla}\mathbf{u}(\mathbf{x}) = -\frac{1}{\ell}\int \mathrm{d}\mathbf{r}\ (\nabla G)_{\ell}(\mathbf{r})[\mathbf{u}(\mathbf{x}+\mathbf{r}) - \mathbf{u}(\mathbf{x})], \tag{33}$$

where an integration by parts has been performed and the condition

$$\int \mathrm{d}\mathbf{r}\, \nabla G(\mathbf{r}) = \mathbf{o} \tag{34}$$

was used, which follows from fast decay of $G(\mathbf{r})$ at infinity. Thus, for some constant $A(\mathbf{x})$,

$$\begin{aligned} |\bar{\nabla}\mathbf{u}(\mathbf{x})| &\le \frac{1}{\ell}\int \mathrm{d}\mathbf{r}\ |(\nabla G)_{\ell}(\mathbf{r})| \cdot |\delta\mathbf{u}(\mathbf{x}; \mathbf{r})| \le A(\mathbf{x})\frac{1}{\ell}\int \mathrm{d}\mathbf{r}\ |(\nabla G)_{\ell}(\mathbf{r})| \cdot |\mathbf{r}|^{\alpha(\mathbf{x})} \\ &= A(\mathbf{x})\ell^{\alpha(\mathbf{x})-1}\int \mathrm{d}\mathbf{r}\, |\nabla G(\mathbf{r})| \cdot |\mathbf{r}|^{\alpha(\mathbf{x})} = \mathrm{O}(\ell^{\alpha(\mathbf{x})-1}) \end{aligned} \tag{35}$$

when $\int \mathrm{d}\mathbf{r}\ |\nabla G(\mathbf{r})| \cdot |\mathbf{r}|^{\alpha(\mathbf{x})} < \infty$. This should be compared with the heuristic estimates in (1). In fact, $\alpha(\mathbf{x})$ in the above bound should, technically, be replaced by $\alpha(\mathbf{x}) - \epsilon$ for any small ϵ, since $\mathbf{u}$ may not be Hölder at $\mathbf{x}$ with exponent $\alpha(\mathbf{x})$ but only with a slightly smaller exponent. Hereafter in this section we assume that $\alpha(\mathbf{x})$ is constant in space, the same for all points $\mathbf{x}$.

[1] The validity of this expectation depends upon the global regularity of the solutions of the Navier–Stokes equation. In 3D, this is still an open problem. See http://www.claymath.org/millennium/Navier-Stokes_Equations/Official_Problem_Description.pdf.

We now wish to determine the contribution to $\bar{\nabla}\mathbf{u}$ from various scales. For this purpose, we introduce a *test filter* $\Gamma(\mathbf{r})$ with the same properties as $G(\mathbf{r})$. For example, $\Gamma(\mathbf{r})$ could be simply $G(\mathbf{r})$ itself. We then define a large-scale field at lengths $>\Delta$ by

$$\mathbf{u}^{>}(\mathbf{x}) = \int \mathrm{d}\mathbf{r}\, \Gamma_{\Delta}(\mathbf{r})\, \mathbf{u}(\mathbf{x}+\mathbf{r}), \tag{36}$$

and a corresponding small-scale field at lengths $<\Delta$ by

$$\mathbf{u}^{<}(\mathbf{x}) = \mathbf{u}(\mathbf{x}) - \mathbf{u}^{>}(\mathbf{x}). \tag{37}$$

We shall employ the convention that the length-scale of the test filter will be denoted Δ (as above) when it is $>\ell$ and we are considering infrared locality. On the other hand, we shall denote the test scale as δ when it is $<\ell$ and we are considering ultraviolet locality.

We first consider infrared locality. The issue is how the magnitude of the contribution to the gradient $\bar{\nabla}\mathbf{u}^{>}$ from $\mathbf{u}^{>}$ compares with that $\bar{\nabla}\mathbf{u}$ from the entire field $\mathbf{u}$. Since the two filter operations with respect to G_ℓ and Γ_Δ commute, it follows that

$$\bar{\nabla}\mathbf{u}^{>}(\mathbf{x}) = \int \mathrm{d}\mathbf{r}\, G_{\ell}(\mathbf{r})\, \nabla\mathbf{u}^{>}(\mathbf{x}+\mathbf{r}). \tag{38}$$

However, for every point $\mathbf{y}$, the analogue of (33) holds:

$$\nabla\mathbf{u}^{>}(\mathbf{y}) = -\frac{1}{\Delta}\int \mathrm{d}\boldsymbol{\rho}\, (\nabla\Gamma)_{\Delta}(\boldsymbol{\rho})\delta\mathbf{u}(\mathbf{y};\boldsymbol{\rho}), \tag{39}$$

This implies $\nabla\mathbf{u}^{>}(\mathbf{y}) = \mathrm{O}(\Delta^{\alpha-1})$ everywhere, exactly as in (35). Here we have used the monofractality of $\mathbf{u}$. Substituting into the integral (38) and using the normalization (4) gives

$$|\bar{\nabla}\mathbf{u}^{>}(\mathbf{x})| = \mathrm{O}(\Delta^{\alpha-1}) \tag{40}$$

for all $\mathbf{x}$. If we assume that the upper bound on $\bar{\nabla}\mathbf{u}$ in (35) holds in fact as an asymptotic scaling result, $\bar{\nabla}\mathbf{u}(\mathbf{x}) \sim (\text{const.})\ell^{\alpha-1}$, then we obtain

$$\frac{|\bar{\nabla}\mathbf{u}^{>}(\mathbf{x})|}{|\nabla\mathbf{u}(\mathbf{x})|} = \mathrm{O}\left(\left(\frac{\ell}{\Delta}\right)^{1-\alpha}\right) = \frac{|\bar{\nabla}\mathbf{u}^{<}(\mathbf{x}) - \bar{\nabla}\mathbf{u}(\mathbf{x})|}{|\nabla\mathbf{u}(\mathbf{x})|}. \tag{41}$$

Notice that the expression on the far right follows from that on the far left by linearity of $\bar{\nabla}\mathbf{u}$ and the definition (37) of $\mathbf{u}^{<}$. The result (41) is the *infrared locality of the large-scale gradient*. It is a rigorous analogue of the heuristic estimate (1) in Section 1. It implies that the largest contribution to $\bar{\nabla}\mathbf{u}$ comes from scales near ℓ when $\alpha < 1$ and that $\bar{\nabla}\mathbf{u}^{>}$ is negligible for $\Delta \gg \ell$. Equivalently, infrared locality means that $\bar{\nabla}\mathbf{u}^{<}(\mathbf{x}) \to \bar{\nabla}\mathbf{u}(\mathbf{x})$ pointwise for $\Delta/\ell \to \infty$, with (41) establishing a minimal rate of convergence $\mathrm{O}((\ell/\Delta)^{1-\alpha})$.

Ultraviolet locality should hold as well, since the filtered gradient is principally large-scale, by construction. To prove it, we substitute directly $\mathbf{u}^{<}$ for $\mathbf{u}$ in (33), getting

$$\bar{\nabla}\mathbf{u}^{<}(\mathbf{x}) = -\frac{1}{\ell}\int \mathrm{d}\mathbf{r}\ (\nabla G)_{\ell}(\mathbf{r})[\mathbf{u}^{<}(\mathbf{x}+\mathbf{r}) - \mathbf{u}^{<}(\mathbf{x})]. \tag{42}$$

However, by its definition $\mathbf{u}^{<}(\mathbf{y}) = -\int \mathrm{d}\boldsymbol{\rho}\, \Gamma_{\delta}(\boldsymbol{\rho})[\mathbf{u}(\mathbf{y}+\boldsymbol{\rho}) - \mathbf{u}(\mathbf{y})]$ for all $\mathbf{y}$, using the normalization condition on Γ. Thus, we see that

$$\mathbf{u}^{<}(\mathbf{y}) = \mathrm{O}(\delta^{\alpha}) \tag{43}$$

by a similar argument as in (35), whenever $\int \mathrm{d}\boldsymbol{\rho}\, |\Gamma(\boldsymbol{\rho})| \cdot |\boldsymbol{\rho}|^{\alpha} < \infty$. Since both terms $\mathbf{u}^{<}(\mathbf{x}+\mathbf{r})$, $\mathbf{u}^{<}(\mathbf{x})$ in (42) can be bounded in this way, we easily obtain that

$$\bar{\nabla}\mathbf{u}^{<}(\mathbf{x}) = \mathrm{O}\left(\frac{\delta^{\alpha}}{\ell}\right) \tag{44}$$

for all $\mathbf{x}$, whenever $\int \mathrm{d}\mathbf{r}\, |\nabla G(\mathbf{r})| < \infty$. Using again the assumption that $\bar{\nabla}\mathbf{u}(\mathbf{x}) \sim (\text{const.})\ell^{\alpha-1}$, we obtain

$$\frac{|\bar{\nabla}\mathbf{u}^{<}(\mathbf{x})|}{|\bar{\nabla}\mathbf{u}(\mathbf{x})|} = \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{\alpha}\right) = \frac{|\bar{\nabla}\mathbf{u}^{>}(\mathbf{x}) - \bar{\nabla}\mathbf{u}(\mathbf{x})|}{|\bar{\nabla}\mathbf{u}(\mathbf{x})|}. \tag{45}$$

This is the statement of *ultraviolet locality of the large-scale gradient*. It implies that $\bar{\nabla}\mathbf{u}^{<}$ is negligible for $\delta/\ell \ll 1$, if $\alpha > 0$, and that $\bar{\nabla}\mathbf{u}^{>}(\mathbf{x}) \to \bar{\nabla}\mathbf{u}(\mathbf{x})$ pointwise for $\delta/\ell \to 0$.

3.1.2. Locality of the small-scale stress

We now establish the locality of $\boldsymbol{\tau}(\mathbf{v},\mathbf{w})$. The key identity that we will use is borrowed from Constantin et al. [16]:

$$\boldsymbol{\tau}(\mathbf{v},\mathbf{w})(\mathbf{x}) = \int \mathrm{d}\mathbf{r}\, G_{\ell}(\mathbf{r})\, \delta\mathbf{v}(\mathbf{x};\mathbf{r})\delta\mathbf{w}(\mathbf{x};\mathbf{r}) - \int \mathrm{d}\mathbf{r}\, G_{\ell}(\mathbf{r})\, \delta\mathbf{v}(\mathbf{x};\mathbf{r}) \int \mathrm{d}\mathbf{r}\, G_{\ell}(\mathbf{r})\, \delta\mathbf{w}(\mathbf{x};\mathbf{r}). \tag{46}$$

The proof is straightforward, by multiplying out the velocity increments and performing the integrations. A very similar identity was proved independently by Vremen et al. [17]. In our earlier work [7] we pointed out that this identity leads to an estimate for $\boldsymbol{\tau}(\mathbf{v},\mathbf{w})$ which is simultaneously local in space and in scale:

$$|\boldsymbol{\tau}(\mathbf{v},\mathbf{w})(\mathbf{x})| = \mathrm{O}(\ell^{\beta(\mathbf{x})+\gamma(\mathbf{x})}) \tag{47}$$

when $\int \mathrm{d}\mathbf{r}\, |G(\mathbf{r})| \cdot |\mathbf{r}|^{p} < \infty$ for each of $p = \beta(\mathbf{x})$, $\gamma(\mathbf{x})$, and $\beta(\mathbf{x}) + \gamma(\mathbf{x})$. Here $\beta(\mathbf{x})$ and $\gamma(\mathbf{x})$ are the pointwise Hölder exponents at location $\mathbf{x}$ of fields $\mathbf{v}$ and $\mathbf{w}$, respectively. Inequality (47) is a precise version of the heuristic estimate in (2) in Section 1 (for the case $\alpha = \beta = \gamma$). Its proof is exactly like that of the similar result proved for $\bar{\nabla}\mathbf{u}$ in (35).

Since the stress is a small-scale quantity, it is more natural to consider first its ultraviolet locality. This is also fairly easy to prove, since the proof is almost the same as that for the large-scale gradient given in the previous subsection. As there, we use the test filter Γ_{δ} to decompose into scales $>\delta$ and $<\delta$. We consider then

$$\boldsymbol{\tau}(\mathbf{v}^{<},\mathbf{w})(\mathbf{x}) = \int \mathrm{d}\mathbf{r}\, G_{\ell}(\mathbf{r})\, \delta\mathbf{v}^{<}(\mathbf{x};\mathbf{r})\delta\mathbf{w}(\mathbf{x};\mathbf{r}) - \int \mathrm{d}\mathbf{r}\, G_{\ell}(\mathbf{r})\, \delta\mathbf{v}^{<}(\mathbf{x};\mathbf{r}) \int \mathrm{d}\mathbf{r}\, G_{\ell}(\mathbf{r})\, \delta\mathbf{w}(\mathbf{x};\mathbf{r}). \tag{48}$$

But using the result that $|\mathbf{v}^{<}(\mathbf{y})| = \mathrm{O}(\delta^{\beta})$ for all $\mathbf{y}$, which was given in (43), we can easily see that $\delta\mathbf{v}^{<}(\mathbf{x};\mathbf{r}) = \mathrm{O}(\delta^{\beta})$. Substituting this estimate into (48), it follows that

$$\boldsymbol{\tau}(\mathbf{v}^{<},\mathbf{w})(\mathbf{x}) = \mathrm{O}(\delta^{\beta}\ell^{\gamma}) \tag{49}$$

now assuming that $\beta(\mathbf{x})$, $\gamma(\mathbf{x})$ are uniform over space. The condition on the filter functions for validity of (49) is $\int \mathrm{d}\mathbf{r}\, |G(\mathbf{r})| \cdot |\mathbf{r}|^{\gamma} < \infty$ and the similar condition for Γ and β. If we then use also (47), assumed to hold in the strong sense of asymptotic scaling and not just a big-O bound, we obtain

$$\frac{|\boldsymbol{\tau}(\mathbf{v}^{<},\mathbf{w})(\mathbf{x})|}{|\boldsymbol{\tau}(\mathbf{v},\mathbf{w})(\mathbf{x})|} = \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{\beta}\right) = \frac{|\boldsymbol{\tau}(\mathbf{v}^{>},\mathbf{w})(\mathbf{x}) - \boldsymbol{\tau}(\mathbf{v},\mathbf{w})(\mathbf{x})|}{|\boldsymbol{\tau}(\mathbf{v},\mathbf{w})(\mathbf{x})|}. \tag{50}$$

This is the statement of *ultraviolet locality of the small-scale stress*. It implies that $\boldsymbol{\tau}(\mathbf{v}^{<},\mathbf{w})$ is negligible for $\delta/\ell \ll 1$, if $\beta > 0$, and that $\boldsymbol{\tau}(\mathbf{v}^{>},\mathbf{w})(\mathbf{x}) \to \boldsymbol{\tau}(\mathbf{v},\mathbf{w})(\mathbf{x})$ pointwise for $\delta/\ell \to 0$. We could just as well have stated the corresponding results for $\boldsymbol{\tau}(\mathbf{v},\mathbf{w}^{<})$ and $\boldsymbol{\tau}(\mathbf{v},\mathbf{w}^{>})$. By symmetry of the bilinear form, these are equivalent to (50).

These results do not correspond to the naive estimate in (2), because there we assumed that *both* of $\mathbf{v}$ and $\mathbf{w}$ were coming from the very small-scales. The analogous rigorous result is for $\boldsymbol{\tau}(\mathbf{v}^<, \mathbf{w}^<)$ and follows from

$$\boldsymbol{\tau}(\mathbf{v}^<, \mathbf{w}^<)(\mathbf{x}) = \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\, \delta\mathbf{v}^<(\mathbf{x};\mathbf{r})\delta\mathbf{w}^<(\mathbf{x};\mathbf{r}) - \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\, \delta\mathbf{v}^<(\mathbf{x};\mathbf{r}) \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\, \delta\mathbf{w}^<(\mathbf{x};\mathbf{r}). \tag{51}$$

If we use in (51) that $|\mathbf{v}^<(\mathbf{y})| = \mathrm{O}(\delta^\beta)$, $|\mathbf{w}^<(\mathbf{y})| = \mathrm{O}(\delta^\gamma)$ for all $\mathbf{y}$, we get

$$\boldsymbol{\tau}(\mathbf{v}^<, \mathbf{w}^<)(\mathbf{x}) = \mathrm{O}(\delta^{\beta+\gamma}). \tag{52}$$

From this we conclude as above that

$$\frac{|\boldsymbol{\tau}(\mathbf{v}^<, \mathbf{w}^<)(\mathbf{x})|}{|\boldsymbol{\tau}(\mathbf{v}, \mathbf{w})(\mathbf{x})|} = \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{\beta+\gamma}\right) \tag{53}$$

This estimate is the exact analogue of (2). It implies that $\boldsymbol{\tau}(\mathbf{v}^<, \mathbf{w}^<)$ is negligible for $\delta/\ell \ll 1$, if $\beta + \gamma > 0$. A corresponding convergence statement follows from (50), (53), and the bilinearity of the $\boldsymbol{\tau}$-form:

$$\frac{|\boldsymbol{\tau}(\mathbf{v}^>, \mathbf{w}^>)(\mathbf{x}) - \boldsymbol{\tau}(\mathbf{v}, \mathbf{w})(\mathbf{x})|}{|\boldsymbol{\tau}(\mathbf{v}, \mathbf{w})(\mathbf{x})|} = \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{\beta}\right) + \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{\gamma}\right) \tag{54}$$

Hence, $\boldsymbol{\tau}(\mathbf{v}^>, \mathbf{w}^>)(\mathbf{x}) \to \boldsymbol{\tau}(\mathbf{v}, \mathbf{w})(\mathbf{x})$ pointwise for $\delta/\ell \to 0$, if both $\beta > 0$ and $\gamma > 0$.

Infrared locality of the stress ought to hold as well, since the stress is naturally a small-scale quantity. To prove it, we need to use the fact that the test-filtered field $\mathbf{v}^>(\mathbf{x})$ is smooth, in the form of a precise estimate on $\delta\mathbf{v}^>(\mathbf{x};\mathbf{r})$. We can get this by writing

$$\delta\mathbf{v}^>(\mathbf{x};\mathbf{r}) = \int \mathrm{d}\boldsymbol{\rho}\, \Gamma_\Delta(\boldsymbol{\rho})[\mathbf{v}(\mathbf{x}+\boldsymbol{\rho}+\mathbf{r}) - \mathbf{v}(\mathbf{x}+\boldsymbol{\rho})] \quad = \int \mathrm{d}\boldsymbol{\rho}\, [\Gamma_\Delta(\boldsymbol{\rho}-\mathbf{x}-\mathbf{r}) - \Gamma_\Delta(\boldsymbol{\rho}-\mathbf{x})]\mathbf{v}(\boldsymbol{\rho}) \tag{55}$$

and using the smoothness of Γ to write

$$\Gamma_\Delta(\boldsymbol{\rho}-\mathbf{x}-\mathbf{r}) - \Gamma_\Delta(\boldsymbol{\rho}-\mathbf{x}) = -\frac{\mathbf{r}}{\Delta} \cdot \int_0^1 \mathrm{d}\theta\, (\nabla\Gamma)_\Delta(\boldsymbol{\rho}-\mathbf{x}-\theta\mathbf{r}). \tag{56}$$

Combining (55) and (56) yields

$$\begin{aligned}\delta\mathbf{v}^>(\mathbf{x};\mathbf{r}) &= -\frac{\mathbf{r}}{\Delta} \cdot \int_0^1 \mathrm{d}\theta \int \mathrm{d}\boldsymbol{\rho}\, (\nabla\Gamma)_\Delta(\boldsymbol{\rho}-\mathbf{x}-\theta\mathbf{r})\mathbf{v}(\boldsymbol{\rho}) \\ &= -\frac{\mathbf{r}}{\Delta} \cdot \int_0^1 \mathrm{d}\theta \int \mathrm{d}\boldsymbol{\rho}\, (\nabla\Gamma)_\Delta(\boldsymbol{\rho}-\mathbf{x}-\theta\mathbf{r})[\mathbf{v}(\boldsymbol{\rho}) - \mathbf{v}(\mathbf{x}+\theta\mathbf{r})]\end{aligned} \tag{57}$$

In the last line we used $\int \mathrm{d}\boldsymbol{\rho}\, \nabla\Gamma(\boldsymbol{\rho}) = \mathbf{0}$ in order to insert the subtracted term. It is now straightforward to derive from (57) that

$$|\delta\mathbf{v}^>(\mathbf{x};\mathbf{r})| = \mathrm{O}(r\Delta^{\beta-1}), \tag{58}$$

for any test filter satisfying $\int \mathrm{d}\boldsymbol{\rho}\, |\nabla\Gamma(\boldsymbol{\rho})| \cdot |\boldsymbol{\rho}|^\beta < \infty$.

With this estimate in hand, we now consider

$$\boldsymbol{\tau}(\mathbf{v}^>, \mathbf{w})(\mathbf{x}) = \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\, \delta\mathbf{v}^>(\mathbf{x};\mathbf{r})\delta\mathbf{w}(\mathbf{x};\mathbf{r}) - \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\, \delta\mathbf{v}^>(\mathbf{x};\mathbf{r}) \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\, \delta\mathbf{w}(\mathbf{x};\mathbf{r}). \tag{59}$$

Repeating the derivation of (47) but using (58) for $\delta\mathbf{v}^>(\mathbf{r})$ rather than the Hölder condition for $\delta\mathbf{v}(\mathbf{r})$, we get that

$$\boldsymbol{\tau}(\mathbf{v}^>, \mathbf{w})(\mathbf{x}) = \mathrm{O}(\Delta^{\beta-1}\ell^{\gamma+1}) \tag{60}$$

when $\int \mathrm{d}\mathbf{r}\,|G(\mathbf{r})|\cdot|\mathbf{r}|^p < \infty$ for each of $p = 1, \gamma$, and $1+\gamma$. Combining with (47), we have finally

$$\frac{|\boldsymbol{\tau}(\mathbf{v}^>, \mathbf{w})(\mathbf{x})|}{|\boldsymbol{\tau}(\mathbf{v}, \mathbf{w})(\mathbf{x})|} = \mathrm{O}\left(\left(\frac{\ell}{\Delta}\right)^{1-\beta}\right) = \frac{|\boldsymbol{\tau}(\mathbf{v}^<, \mathbf{w})(\mathbf{x}) - \boldsymbol{\tau}(\mathbf{v}, \mathbf{w})(\mathbf{x})|}{|\boldsymbol{\tau}(\mathbf{v}, \mathbf{w})(\mathbf{x})|}. \tag{61}$$

This is the statement of *infrared locality of the small-scale stress*. It implies that $\boldsymbol{\tau}(\mathbf{v}^>, \mathbf{w})$ is negligible for $\Delta/\ell \gg 1$, if $\beta < 1$, and that $\boldsymbol{\tau}(\mathbf{v}^<, \mathbf{w})(\mathbf{x}) \to \boldsymbol{\tau}(\mathbf{v}, \mathbf{w})(\mathbf{x})$ pointwise for $\Delta/\ell \to \infty$. Again, we could just as well have stated the equivalent results for $\boldsymbol{\tau}(\mathbf{v}, \mathbf{w}^>)$ and $\boldsymbol{\tau}(\mathbf{v}, \mathbf{w}^<)$.

Of course, one can also get similar but stronger bounds if *both* of $\mathbf{v}$ and $\mathbf{w}$ are replaced by their large-scale parts. Considering

$$\boldsymbol{\tau}(\mathbf{v}^>, \mathbf{w}^>)(\mathbf{x}) = \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\,\delta\mathbf{v}^>(\mathbf{x};\mathbf{r})\delta\mathbf{w}^>(\mathbf{x};\mathbf{r}) - \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\,\delta\mathbf{v}^>(\mathbf{x};\mathbf{r}) \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\,\delta\mathbf{w}^>(\mathbf{x};\mathbf{r}) \tag{62}$$

and repeating the arguments leading to (60) we get

$$\boldsymbol{\tau}(\mathbf{v}^>, \mathbf{w}^>)(\mathbf{x}) = \mathrm{O}(\Delta^{\beta+\gamma-2}\ell^2) \tag{63}$$

for any filter functions satisfying $\int \mathrm{d}\boldsymbol{\rho}\,|\nabla\Gamma(\boldsymbol{\rho})|\cdot|\boldsymbol{\rho}|^p < \infty$ for $p = \beta, \gamma$ and $\int \mathrm{d}\mathbf{r}\,|G(\mathbf{r})|\cdot|\mathbf{r}|^p < \infty$ for $p = 1, 2$. Thus, we obtain

$$\frac{|\boldsymbol{\tau}(\mathbf{v}^>, \mathbf{w}^>)(\mathbf{x})|}{|\boldsymbol{\tau}(\mathbf{v}, \mathbf{w})(\mathbf{x})|} = \mathrm{O}\left(\left(\frac{\ell}{\Delta}\right)^{2-\beta-\gamma}\right) \tag{64}$$

which is negligible for $\Delta/\ell \gg 1$ if $\beta + \gamma < 2$. Furthermore, there is also the result

$$\frac{|\boldsymbol{\tau}(\mathbf{v}^<, \mathbf{w}^<)(\mathbf{x}) - \boldsymbol{\tau}(\mathbf{v}, \mathbf{w})(\mathbf{x})|}{|\boldsymbol{\tau}(\mathbf{v}, \mathbf{w})(\mathbf{x})|} = \mathrm{O}\left(\left(\frac{\ell}{\Delta}\right)^{1-\beta}\right) + \mathrm{O}\left(\left(\frac{\ell}{\Delta}\right)^{1-\gamma}\right) \tag{65}$$

which follows from (61), (64) and bilinearity of the $\boldsymbol{\tau}$-form. Thus $\boldsymbol{\tau}(\mathbf{v}^<, \mathbf{w}^<)(\mathbf{x}) \to \boldsymbol{\tau}(\mathbf{v}, \mathbf{w})(\mathbf{x})$ pointwise for $\Delta/\ell \to \infty$, if both $\beta < 1$ and $\gamma < 1$.

3.1.3. Locality of the flux

The results in the previous two sections yield similar statements for the fluxes $\Pi(\mathbf{u}, \mathbf{v}, \mathbf{w})$. For example, there is infrared locality for $\Delta > \ell$:

$$\frac{|\Pi(\mathbf{u}^>, \mathbf{v}, \mathbf{w})(\mathbf{x})|}{|\Pi(\mathbf{u}, \mathbf{v}, \mathbf{w})(\mathbf{x})|} = \mathrm{O}\left(\left(\frac{\ell}{\Delta}\right)^{1-\alpha}\right) \tag{66}$$

and ultraviolet locality for $\delta < \ell$:

$$\frac{|\Pi(\mathbf{u}^<, \mathbf{v}, \mathbf{w})(\mathbf{x})|}{|\Pi(\mathbf{u}, \mathbf{v}, \mathbf{w})(\mathbf{x})|} = \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{\alpha}\right) \tag{67}$$

Rather than replacing $\mathbf{u}$ by $\mathbf{u}^>$ or $\mathbf{u}^<$, one could also replace $\mathbf{v}$ by $\mathbf{v}^>$ or $\mathbf{v}^<$, and statements corresponding to (66) and (67) would hold with β appearing rather than α. The same statements are true also with $\mathbf{w}$ and γ. Replacements of two or three fields yield smaller estimates, by a corresponding number of factors, one for each replacement. These estimates can be restated as convergence results, in an obvious way. It is interesting that infrared locality in a particular field, say, $\mathbf{u}$, is always determined by the condition for its corresponding Hölder exponent that $\alpha < 1$. Likewise, ultraviolet locality in a particular field, say, again, $\mathbf{u}$, is determined by the condition that its exponent $\alpha > 0$. Thus, locality in both directions is guaranteed when all the fields involved are, roughly speaking, continuous but non-differentiable.

3.2. Multifractal fields

In the entire preceding discussion we have assumed that the fields involved are all monofractal. However, monofractality is not expected to be necessary for locality of transfer, i.e. the locality property does not imply monofractality. As an illustrative example, consider so-called "shell models" of turbulence (e.g. see [18], Section 8.7). These are toy dynamical models of turbulence that, by construction, are strongly local, with interactions only between neighboring shells. Nevertheless, these models exhibit anomalous scaling of structure functions, or scaling exponents ζ_p of pth-order moments which are not linear functions of the order [19–21]. Such scaling has been interpreted in terms of a "multifractal model" for the velocity field, with Hölder exponents that vary from point to point in space [2] (or, for the shell model, in time). Whether the "multifractal model" is rigorously implied by anomalous scaling and the precise sense of its validity is still an open question; see [14] for a recent mathematical discussion. However, it is not hard to show at least that a random velocity field with such anomalous scaling will have almost sure local Hölder regularity and the exponents will depend, in general, upon the space-point considered [4,14]. Thus, it is important to establish conditions for locality in this generality.

In fact, much of our analysis seems to be local in space and would appear to apply to a general multifractal field. Therefore, we must explain some of the essential difficulties to extend to that case. Consider, for example, the proof of the infrared locality of $\bar{\nabla}\mathbf{u}$ in Section 3.1.1. Let us assume, for simplicity, that the filter function G is compactly supported in the unit ball. The formula (38) together with $\nabla\mathbf{u}^{>}(\mathbf{x}+\mathbf{r}) = \mathrm{O}(\Delta^{\alpha(\mathbf{x}+\mathbf{r})-1})$ then yields a local result

$$\bar{\nabla}\mathbf{u}^{>}(\mathbf{x}) = \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\, \nabla\mathbf{u}^{>}(\mathbf{x}+\mathbf{r}) = \mathrm{O}(\Delta^{\alpha_*(\mathbf{x};\,\ell)-1}) \tag{68}$$

where

$$\alpha_*(\mathbf{x};\ell) = \inf_{|\mathbf{r}|<\ell} \alpha(\mathbf{x}+\mathbf{r}). \tag{69}$$

Thus, the estimate must take into account the *worst* singularity in the neighborhood of the point $\mathbf{x}$. Notice that as $\ell \to 0$, the function $\alpha_*(\mathbf{x};\ell) \nearrow \alpha_*(\mathbf{x})$ where $\alpha_*(\mathbf{x}) \le \alpha(\mathbf{x})$ is a lower semicontinuous function, the lower envelope function of the pointwise Hölder exponent $\alpha(\mathbf{x})$. This function $\alpha_*(\mathbf{x})$ is called the *local Hölder exponent* of $\mathbf{u}$ at point $\mathbf{x}$ [22]. Thus, by taking ℓ sufficiently small, we can replace $\alpha_*(\mathbf{x};\,\ell)$ in (68) by $\alpha_*(\mathbf{x}) - \epsilon$, with $\epsilon > 0$ arbitrarily small. This gives an apparently local estimate. However, that conclusion is deceptive, since generally for multifractal functions $\alpha_*(\mathbf{x})$ is a *constant* equal to $\alpha_{\min}$, the global minimum Hölder exponent [22]. The reason is that, at least for typical multifractal functions that have been rigorously studied, the sets $\mathcal{S}_\alpha = \{\mathbf{x} : \alpha(\mathbf{x}) = \alpha\}$ are dense (or else empty) for all α. Thus, the worst singularity appears in every open ball and dominates in an estimate like (68). Another related difficulty is that our assumptions of *pointwise* asymptotic scaling, e.g.

$$\bar{\nabla}\mathbf{u}(\mathbf{x}) \sim (\text{const.})\ell^{\alpha(\mathbf{x})-1}, \tag{70}$$

that was made throughout Section 3.1 is unlikely to hold, even for monofractal functions. In general the velocity increment $\delta\mathbf{u}(\mathbf{x};\mathbf{r})$ scales in a highly oscillatory manner as $r \to 0$ [24,25], and this is likely to be true also for $\bar{\nabla}\mathbf{u}(\mathbf{x})$.

An approach which avoids these difficulties is to consider scale locality in the space-mean sense, rather than pointwise scaling. Thus, from the physical point of view, we consider the *structure–function scaling exponents* rather than local Hölder exponents. These can be defined for $p \ge 1$ by taking first $\nu \to 0$ and then

$$\alpha_p = \liminf_{\mathbf{r}\to\mathbf{O}} \frac{\log \|\delta\mathbf{u}(\mathbf{r})\|_p}{\log r} \tag{71}$$

Here $\|\cdot\|_p$ denotes the standard L^p-norm on the space domain of the flow. In more mathematical language, the exponents in (71) are the *maximal Besov indices* [4]. These are related to the usual (absolute) structure–function exponents by $\alpha_p = \zeta_p/p$ and are thus decreasing functions of p by concavity of ζ_p (or directly by monotonicity

of the L_p-norms in p.) In [7] we used these exponents to establish a bound on the space-average pth moments or L_p-norms of the local energy flux. A generalization of that result states that

$$\|\Pi(\mathbf{u},\mathbf{v},\mathbf{w})\|_p = \mathrm{O}(\ell^{\alpha_q+\beta_r+\gamma_s-1}) \tag{72}$$

where $q, r, s \geq 1$ satisfy $(1/q)+(1/r)+(1/s)=(1/p)$ and $\alpha_q, \beta_r, \gamma_s$ are the corresponding scaling exponents of the fields $\mathbf{u}, \mathbf{v}, \mathbf{w}$. This bound is optimized with

$$\mu_p = \sup\left\{\alpha_q+\beta_r+\gamma_s-1 : \frac{1}{q}+\frac{1}{r}+\frac{1}{s}=\frac{1}{p}\right\} \quad = \alpha_{q_p^*}+\beta_{r_p^*}+\gamma_{s_p^*}-1 \tag{73}$$

Thus, $\|\Pi(\mathbf{u},\mathbf{v},\mathbf{w})\|_p = \mathrm{O}(\ell^{\mu_p})$. In fact, we might expect that asymptotic scaling holds:

$$\|\Pi(\mathbf{u},\mathbf{v},\mathbf{w})\|_p \sim (\text{const.})\ell^{\mu_p}. \tag{74}$$

A special case of this scaling law was proposed already by Kraichnan [23] for the 3D energy cascade when $\mathbf{u}=\mathbf{v}=\mathbf{w}$ all equal the fluid velocity. As an alternative to Kolmogorov's "refined similarity hypothesis," he suggested that the scaling exponents $\zeta_p = p\alpha_p$ would be related to those $\tau_p = p\mu_p$ of the local energy flux by

$$\zeta_p = \frac{p}{3}+\tau_{p/3}. \tag{75}$$

The relation (75) has been well verified in numerical simulations [26,27]. Our proposal (73) generalizes that of Kraichnan, since in his case $\beta_r=\alpha_r$, $\gamma_s=\alpha_s$. Thus, the supremum in (73) must, if uniquely achieved, be at $q_p^*=r_p^*=s_p^*=3p$, by symmetry. This yields $\mu_p = 3\alpha_{3p}-1$, equivalent to (75).

Before establishing locality of cascades in the space-mean sense, we should first give a brief proof of (72). Let us start with $\bar{\nabla}\mathbf{u}$. Using (33) and the triangle inequality

$$\|\bar{\nabla}\mathbf{u}\|_q \leq \frac{1}{\ell}\int \mathrm{d}\mathbf{r}\ |(\nabla G)_\ell(\mathbf{r})|\cdot\|\delta\mathbf{u}(\mathbf{r})\|_q = \mathrm{O}(\ell^{\alpha_q-1}). \tag{76}$$

Next consider $\boldsymbol{\tau}(\mathbf{v},\mathbf{w})$. We shall use (46), written as

$$\boldsymbol{\tau}(\mathbf{v},\mathbf{w})(\mathbf{x}) = \boldsymbol{\rho}(\mathbf{v},\mathbf{w})(\mathbf{x}) - \mathbf{v}'(\mathbf{x})\mathbf{w}'(\mathbf{x}) \tag{77}$$

with

$$\boldsymbol{\rho}(\mathbf{v},\mathbf{w})(\mathbf{x}) = \int \mathrm{d}\mathbf{r}\ G_\ell(\mathbf{r})\,\delta\mathbf{v}(\mathbf{x};\mathbf{r})\delta\mathbf{w}(\mathbf{x};\mathbf{r}) \tag{78}$$

and

$$\mathbf{v}'(\mathbf{x}) = \mathbf{v}(\mathbf{x}) - \bar{\mathbf{v}}(\mathbf{x}) = -\int \mathrm{d}\mathbf{r}\ G_\ell(\mathbf{r})\,\delta\mathbf{v}(\mathbf{x};\mathbf{r}). \tag{79}$$

If we define t satisfying $(1/q)+(1/t)=(1/p)$ and $(1/r)+(1/s)=(1/t)$, then by the triangle and Hölder inequalities

$$\|\boldsymbol{\rho}(\mathbf{v},\mathbf{w})\|_t \leq \int \mathrm{d}\mathbf{r}\ G_\ell(\mathbf{r})\,\|\delta\mathbf{v}(\mathbf{r})\delta\mathbf{w}(\mathbf{r})\|_t \leq \int \mathrm{d}\mathbf{r}\ G_\ell(\mathbf{r})\,\|\delta\mathbf{v}(\mathbf{r})\|_r\,\|\delta\mathbf{w}(\mathbf{r})\|_s = \mathrm{O}(\ell^{\beta_r+\gamma_s}) \tag{80}$$

But also from (79) and the triangle inequality

$$\|\mathbf{v}'\|_r \leq \int \mathrm{d}\mathbf{r}\ G_\ell(\mathbf{r})\,\|\delta\mathbf{v}(\mathbf{r})\|_r = \mathrm{O}(\ell^{\beta_r}) \tag{81}$$

Likewise, $\|\mathbf{w}'\|_s = \mathrm{O}(\ell^{\gamma_s})$. Putting together (80) and (81), it follows that

$$\|\boldsymbol{\tau}(\mathbf{v},\mathbf{w})\|_t = \mathrm{O}(\ell^{\beta_r+\gamma_s}). \tag{82}$$

Finally, using (76), (82), and the Hölder inequality, one gets (72). The conditions on the filter function G necessary to derive this estimate are $\int \mathrm{d}\mathbf{r}\,|\nabla G(\mathbf{r})|\,|\mathbf{r}|^{\nu} < \infty$ for $\nu = \alpha_q$ and $\int \mathrm{d}\mathbf{r}\,|G(\mathbf{r})|\,|\mathbf{r}|^{\nu} < \infty$ for $\nu = \beta_r$, γ_s, and $\beta_r + \gamma_s$.

The estimate (81) is classical. Equivalently, we may state it as

$$\|\mathbf{v} - \bar{\mathbf{v}}\|_r = \mathrm{O}(\ell^{\beta})$$

if $\mathbf{v} \in B_r^{\beta}$. Whenever this holds, then $\bar{\mathbf{v}} \to \mathbf{v}$ in the L^r-sense as $\ell \to 0$, with a rate determined by $\beta > 0$. In fact, this approximation property characterizes the Besov space B_r^{β}, e.g. see [28], Section 3.7. As this remark illustrates, we can still obtain useful convergence results in the space-mean setting.

We shall now establish successively the scale-locality of the large-scale gradient, the stress, and the flux, in the space-mean sense.

3.2.1. Locality of the large-scale gradient

Let us first discuss infrared locality for the large-scale gradient. As for the pointwise estimate, we use (38) but now with the triangle inequality to get

$$\|\bar{\nabla}\mathbf{u}^{>}\|_q \le \int \mathrm{d}\mathbf{r}\, G_{\ell}(\mathbf{r})\, \|\nabla\mathbf{u}^{>}\|_q = \mathrm{O}(\Delta^{\alpha_q - 1}). \tag{83}$$

The last statement holds since $\|\nabla\mathbf{u}^{>}\|_q = \mathrm{O}(\Delta^{\alpha_q-1})$ by (76). Thus, if

$$\|\bar{\nabla}\mathbf{u}\|_q \sim (\text{const.})\ell^{\alpha_q - 1}, \tag{84}$$

then

$$\frac{\|\bar{\nabla}\mathbf{u}^{>}\|_q}{\|\bar{\nabla}\mathbf{u}\|_q} = \mathrm{O}\left(\left(\frac{\ell}{\Delta}\right)^{1-\alpha_q}\right) = \frac{\|\bar{\nabla}\mathbf{u}^{<} - \bar{\nabla}\mathbf{u}\|_q}{\|\bar{\nabla}\mathbf{u}\|_q}. \tag{85}$$

The result (85) expresses the *infrared locality of the large-scale gradient in the L^q space-mean sense*, for any $\alpha_q < 1$.

This is an appropriate point at which to remark that we do not need the strict scaling relation (84) to hold in order to derive useful results about scale locality. Suppose that instead

$$\|\bar{\nabla}\mathbf{u}\|_q \ge (\text{const.})\ell^{\alpha'_q - 1}, \tag{86}$$

for all $\ell < \ell_0$ with some $\alpha'_q > \alpha_q$. For example, it might be the case that the asymptotic scaling relation $\|\bar{\nabla}\mathbf{u}\|_q \sim (\text{const.})\ell^{\alpha'_q - 1}$ holds as $\ell \to 0$ with some $\alpha'_q > \alpha_q$. Then

$$\frac{\|\bar{\nabla}\mathbf{u}^{>}\|_q}{\|\bar{\nabla}\mathbf{u}\|_q} = \mathrm{O}\left(\left(\frac{\ell}{\Delta}\right)^{1-\alpha_q}\left(\frac{L}{\ell}\right)^{\alpha'_q - \alpha_q}\right) = \frac{\|\bar{\nabla}\mathbf{u}^{<} - \bar{\nabla}\mathbf{u}\|_q}{\|\bar{\nabla}\mathbf{u}\|_q}, \tag{87}$$

where L is a fixed length of the order of the integral scale. We still get an estimate going to zero for $\Delta/\ell \to \infty$ as long as $\alpha'_q < 1$, but at a somewhat reduced rate. For example, if $\Delta = L$, then the bound is $\mathrm{O}((\ell/L)^{1-\alpha'_q})$ not $\mathrm{O}((\ell/L)^{1-\alpha_q})$. Similar statements apply to all of our previous and subsequent locality estimates. Although we shall state them under the simplest scaling assumption, analogous to (84), they all have simple modifications for weakened hypotheses corresponding to (86). In particular, our scaling hypothesis (74) for flux could hold instead with some $\mu'_p > \mu_p$ and scale-locality of cascade would still follow.

We now establish ultraviolet locality for the large-scale gradient. As for the pointwise estimate, we use (42) but now with the triangle inequality to get

$$\|\bar{\nabla}\mathbf{u}^<\|_q \le \frac{2}{\ell}\int \mathrm{d}\mathbf{r}\,|(\nabla G)_\ell(\mathbf{r})|\cdot\|\mathbf{u}^<\|_q = \mathrm{O}\left(\frac{\delta^{\alpha_q}}{\ell}\right). \tag{88}$$

The last line follows since $\|\mathbf{u}^<\|_q = \mathrm{O}(\delta^{\alpha_q})$ from (81). Thus if we assume that $\|\bar{\nabla}\mathbf{u}\|_q \sim (\text{const.})\ell^{\alpha_q-1}$, then we obtain

$$\frac{\|\bar{\nabla}\mathbf{u}^<\|_q}{\|\bar{\nabla}\mathbf{u}\|_q} = \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{\alpha_q}\right) = \frac{\|\bar{\nabla}\mathbf{u}^> - \bar{\nabla}\mathbf{u}\|_q}{\|\bar{\nabla}\mathbf{u}\|_q}. \tag{89}$$

This result expresses the *ultraviolet locality of the large-scale gradient in the L^q space-mean sense* when $\alpha_q > 0$. The filter functions must only satisfy $\int \mathrm{d}\mathbf{r}\,|\nabla G(\mathbf{r})| < \infty$ and $\int \mathrm{d}\boldsymbol{\rho}\,|\Gamma(\boldsymbol{\rho})|\cdot\rho^{\alpha_q} < \infty$ for (89) to hold. Later on, in Section 4.2, we shall show that this estimate can be improved in an essential way, if additional smoothness conditions on the filter functions are imposed.

3.2.2. Locality of the small-scale stress

We now turn to the locality of the stress $\boldsymbol{\tau}(\mathbf{v},\mathbf{w})$. First we consider the ultraviolet. As for the pointwise estimates, we use (48) but now with the triangle and Hölder inequalities to get

$$\|\boldsymbol{\tau}(\mathbf{v}^<,\mathbf{w})\|_t \le 2\int \mathrm{d}\mathbf{r}\,G_\ell(\mathbf{r})\,\|\mathbf{v}^<\|_r\|\delta\mathbf{w}(\mathbf{r})\|_s + 2\int \mathrm{d}\mathbf{r}\,G_\ell(\mathbf{r})\,\|\mathbf{v}^<\|_r\int \mathrm{d}\mathbf{r}\,G_\ell(\mathbf{r})\,\|\delta\mathbf{w}(\mathbf{r})\|_s \tag{90}$$

for any $r, s, t \ge 1$ satisfying $(1/r) + (1/s) = (1/t)$. Then, since $\|\mathbf{v}^<\|_r = \mathrm{O}(\delta^{\beta_r})$ from (81),

$$\|\boldsymbol{\tau}(\mathbf{v}^<,\mathbf{w})\|_t = \mathrm{O}(\delta^{\beta_r}\ell^{\gamma_s}). \tag{91}$$

As with the flux, we may expect that

$$\|\boldsymbol{\tau}(\mathbf{v},\mathbf{w})\|_t \sim (\text{const.})\ell^{\nu_t} \tag{92}$$

with

$$\nu_t = \sup\left\{\beta_r + \gamma_s : \frac{1}{r} + \frac{1}{s} = \frac{1}{t}\right\} = \beta_{\bar{r}_t} + \gamma_{\bar{s}_t} \tag{93}$$

For $\mathbf{v} = \mathbf{w}$ fluid velocity fields in 3D turbulence it has been found from experiment that this relation holds well with, of course, $\bar{r}_t = \bar{s}_t = 2t$ [29]. It follows that

$$\frac{\|\boldsymbol{\tau}(\mathbf{v}^<,\mathbf{w})\|_t}{\|\boldsymbol{\tau}(\mathbf{v},\mathbf{w})\|_t} = \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{\beta_{\bar{r}_t}}\right) = \frac{\|\boldsymbol{\tau}(\mathbf{v}^>,\mathbf{w}) - \boldsymbol{\tau}(\mathbf{v},\mathbf{w})\|_t}{\|\boldsymbol{\tau}(\mathbf{v},\mathbf{w})\|_t}. \tag{94}$$

This estimate expresses the *ultraviolet locality of the small-scale stress in the L^t space-mean sense* if $\beta_{\bar{r}_t} > 0$. Of course, we can also get space-mean results analogous to the pointwise estimates (53) and (54), in which both fields are replaced by their very small-scale parts. Thus,

$$\frac{\|\boldsymbol{\tau}(\mathbf{v}^<,\mathbf{w}^<)\|_t}{\|\boldsymbol{\tau}(\mathbf{v},\mathbf{w})\|_t} = \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{\nu_t}\right) \tag{95}$$

and

$$\frac{\|\boldsymbol{\tau}(\mathbf{v}^>,\mathbf{w}^>) - \boldsymbol{\tau}(\mathbf{v},\mathbf{w})\|_t}{\|\boldsymbol{\tau}(\mathbf{v},\mathbf{w})\|_t} = \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{\beta_{\bar{r}_t}}\right) + \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{\gamma_{\bar{s}_t}}\right) \tag{96}$$

We leave the proof of these as an easy exercise for the reader.

Infrared locality follows from (57), just as for the pointwise estimate, but uses the triangle inequality to get

$$\|\delta\mathbf{v}^{>}(\mathbf{r})\|_r \le \frac{r}{\Delta}\int_0^1 \mathrm{d}\theta \int \mathrm{d}\boldsymbol{\rho}\, |(\nabla\Gamma)_\Delta(\boldsymbol{\rho})| \cdot \|\delta\mathbf{v}(\boldsymbol{\rho})\|_r \quad = \mathrm{O}(r\Delta^{\beta_r-1}) \tag{97}$$

Using this estimate and the Hölder inequality in (59)

$$\begin{aligned}\|\boldsymbol{\tau}(\mathbf{v}^{>},\mathbf{w})\|_t &\le \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\, \|\delta\mathbf{v}^{>}(\mathbf{r})\|_r\, \|\delta\mathbf{w}(\mathbf{r})\|_s + \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\, \|\delta\mathbf{v}^{>}(\mathbf{r})\|_r \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})\, \|\delta\mathbf{w}(\mathbf{r})\|_s \\ &= \mathrm{O}(\Delta^{\beta_r-1}\ell^{\gamma_s+1})\end{aligned} \tag{98}$$

Combining with (92), we have finally

$$\frac{\|\boldsymbol{\tau}(\mathbf{v}^{>},\mathbf{w})\|_t}{\|\boldsymbol{\tau}(\mathbf{v},\mathbf{w})\|_t} = \mathrm{O}\left(\left(\frac{\ell}{\Delta}\right)^{1-\beta_{\bar{r}_t}}\right) = \frac{\|\boldsymbol{\tau}(\mathbf{v}^{<},\mathbf{w}) - \boldsymbol{\tau}(\mathbf{v},\mathbf{w})\|_t}{\|\boldsymbol{\tau}(\mathbf{v},\mathbf{w})\|_t}. \tag{99}$$

This expresses the *infrared locality of the small-scale stress in the L^t space-mean sense* if $\beta_{\bar{r}_t} < 1$. Of course, results hold analogous to the pointwise estimates (64) and (65) with both fields replaced by large-scale/small-scale parts. We leave the statements and proofs to the reader.

3.2.3. Locality of the flux

Similar results hold for the fluxes. Assuming that (74) holds, we get infrared locality for $\Delta > \ell$,

$$\frac{\|\Pi(\mathbf{u}^{>},\mathbf{v},\mathbf{w})\|_p}{\|\Pi(\mathbf{u},\mathbf{v},\mathbf{w})\|_p} = \mathrm{O}\left(\left(\frac{\ell}{\Delta}\right)^{1-\alpha_{q_p^*}}\right) \tag{100}$$

if $\alpha_{q_p^*} < 1$, and similar statements with $\mathbf{v}^{>}$, $\beta_{r_p^*}$ or $\mathbf{w}^{>}$, $\gamma_{s_p^*}$ in place of $\mathbf{u}^{>}$, $\alpha_{q_p^*}$. Likewise, there is ultraviolet locality for $\delta < \ell$

$$\frac{\|\Pi(\mathbf{u}^{<},\mathbf{v},\mathbf{w})\|_p}{\|\Pi(\mathbf{u},\mathbf{v},\mathbf{w})\|_p} = \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{\alpha_{q_p^*}}\right) \tag{101}$$

if $\alpha_{q_p^*} > 0$, Of course, we can get similar results as those discussed in Section 3.1.3 on pointwise estimation, with either $\mathbf{v}$ or $\mathbf{w}$ replaced or with two or three fields replaced by the very large- or small-scale fields. All these results should now be easy for the reader to state and to prove.

3.3. Summary

Since this section has been lengthy and its contents sometimes a bit technical, it is worth summarizing the conclusions in a more physical language. The basic idea is the same as the heuristic argument that was given in Section 1. In addition, key estimates were developed for the UV and IR contributions to velocity-increments, $\delta u^{<}(\ell)$ and $\delta u^{>}(\ell)$. The basic estimate for $\delta u^{<}(\ell)$ is

$$\delta u^{<}(\ell) \sim \delta^{\alpha}, \tag{102}$$

whose rigorous versions are (43) in the local form and (81) in the global, space-mean sense. Likewise, the basic estimate for $\delta u^{>}(\ell)$ is

$$\delta u^{>}(\ell) \sim \ell\Delta^{\alpha-1}, \tag{103}$$

proved rigorously in (58) locally and in (97) globally.

As claimed in Section 1, the strain at length-scale ℓ scales like

$$S(\ell) \sim \frac{\delta u(\ell)}{\ell} \sim \ell^{\alpha-1}; \tag{104}$$

see Eqs. (35) [local] and (76) [global]. Thus,

$$S^{<}(\ell) \sim \frac{\delta u^{<}(\ell)}{\ell} \sim \frac{\delta^{\alpha}}{\ell}, \tag{105}$$

whence UV locality of strain follows as

$$\frac{S^{<}(\ell)}{S(\ell)} \sim \left(\frac{\delta}{\ell}\right)^{\alpha} \tag{106}$$

when $\alpha > 0$ [cf. (45) local; (89) global]. Also,

$$S^{>}(\ell) \sim \frac{\delta u^{>}(\ell)}{\ell} \sim \Delta^{\alpha-1}, \tag{107}$$

from which IR locality of strain follows as

$$\frac{S^{>}(\ell)}{S(\ell)} \sim \left(\frac{\ell}{\Delta}\right)^{1-\alpha} \tag{108}$$

when $\alpha < 1$ [cf. (41) local; (85) global].

In the same way, the stress at length-scale ℓ scales like

$$\tau(\ell) \sim (\delta u(\ell))^2 \sim \ell^{2\alpha}; \tag{109}$$

see Eqs. (47) [local] and (82) [global]. Then

$$\tau^{<}(\ell) \sim (\delta u^{<}(\ell))^2 \sim \delta^{2\alpha}, \tag{110}$$

from which UV locality of stress follows as

$$\frac{\tau^{<}(\ell)}{\tau(\ell)} \sim \left(\frac{\delta}{\ell}\right)^{2\alpha} \tag{111}$$

when $\alpha > 0$ [cf. (50) local; (94) global]. Finally,

$$\tau^{>}(\ell) \sim (\delta u^{>}(\ell))^2 \sim \ell^2 \Delta^{2\alpha-2}, \tag{112}$$

from which IR locality of stress follows as

$$\frac{\tau^{>}(\ell)}{\tau(\ell)} \sim \left(\frac{\ell}{\Delta}\right)^{2(1-\alpha)} \tag{113}$$

when $\alpha < 1$ [cf. (61) local; (99) global].

4. Three physical examples

We now consider three examples, both for their intrinsic interest, and for the additional light they shed upon our general estimates above.

4.1. Energy cascades in 2D and 3D

Energy cascades in 2D and 3D are an example of our analysis with $\Pi_E = \Pi(\mathbf{u}, \mathbf{u}, \mathbf{u})$, where $\mathbf{u}$ is the fluid velocity. According to the original predictions of Kolmogorov [30–32], Onsager [1,33] and others for 3D, the fluid velocity in the high Reynolds number limit has $\alpha_p = 1/3$ for all $p \geq 1$. The same prediction was made by Kraichnan [34] for the 2D inverse energy cascade. While experiment [35] and simulation [36,37] seem to verify this prediction for the 2D velocity field – which is thus monofractal – the velocity field in 3D appears to be multifractal. Estimates from experiment and simulation in 3D give $\alpha_2 \doteq 0.355$, $\alpha_4 \doteq 0.313$, $\alpha_6 \doteq 0.272$, $\alpha_8 \doteq 0.234$ [38]. Note that we have used here the measurements of scaling exponents for *transverse* velocity-increments, since they seem to be smaller than those for longitudinal increments. This corresponds to our definition (71). Since these exponents all lie between 0 and 1 we get locality of energy cascade in both the infrared and the ultraviolet. For example,

$$\frac{\|\Pi(\mathbf{u}^>, \mathbf{u}, \mathbf{u})\|_1}{\|\Pi(\mathbf{u}, \mathbf{u}, \mathbf{u})\|_1} = \mathrm{O}\left(\left(\frac{\ell}{\Delta}\right)^{1-\alpha_3}\right), \quad \Delta > \ell \tag{114}$$

is an L^1 bound on the amount of energy flux from strain at very large scales. Likewise,

$$\frac{\|\Pi(\mathbf{u}, \mathbf{u}^<, \mathbf{u}^<)\|_1}{\|\Pi(\mathbf{u}, \mathbf{u}, \mathbf{u})\|_1} = \mathrm{O}\left(\left(\frac{\delta}{\ell}\right)^{2\alpha_3}\right), \quad \delta < \ell \tag{115}$$

is an L^1 bound on the amount of energy flux from stress at very small scales. In both 2D and 3D, $\alpha_3 \doteq 1/3$. It is interesting that 1/3 is the special value for which the infrared and ultraviolet bounds decay at the same rate. This property holds also for Kolmogorov spectra in weak turbulence, where it has been called "counterbalanced locality" [39], and for analytical turbulence closures of the DIA-class, as we discuss more below (Section 5). It is unclear what is the significance of the coincidence that this "counterbalance" occurs uniquely for the K41 exponent-value $\alpha_3 = 1/3$. For reasons discussed by Kraichnan [40,41], the 2D cascade is less local than the 3D cascade, although the *asymptotic* rate of decay of nonlocal contributions is the same in both dimensions. L^p-bounds for $p > 1$ can be formulated as well and, in 3D, for increasing p these bounds improve in the infrared and deteriorate in the ultraviolet.

We note here that it is crucial that terms such as $(e + \overline{p})\overline{\mathbf{u}}$ and $\overline{\mathbf{u}} \cdot \boldsymbol{\tau}$ proportional to $\overline{\mathbf{u}}$ can be isolated in the energy transport vector $\mathbf{J}_E$ defined in (12). The velocity *gradients* from larger scales are bounded by $\overline{\nabla}\mathbf{u}^> = \mathrm{O}(\Delta^{\alpha-1})$ and thus get smaller with increasing Δ. (Here α could be either a pointwise or a space-mean scaling exponent.) However,

$$\overline{\mathbf{u}^>}(\mathbf{x}) - \mathbf{u}^>(\mathbf{x}) = \int \mathrm{d}\mathbf{r}\, G_\ell(\mathbf{r})[\mathbf{u}^>(\mathbf{x}+\mathbf{r}) - \mathbf{u}^>(\mathbf{x})] \quad = \int \mathrm{d}\mathbf{r} \int_0^1 \mathrm{d}\theta\, G_\ell(\mathbf{r})\, \mathbf{r} \cdot \nabla \mathbf{u}^>(\mathbf{x}+\theta\mathbf{r}) \quad = \mathrm{O}(\ell\Delta^{\alpha-1}). \tag{116}$$

and thus

$$\overline{\mathbf{u}^>} = \mathbf{u}^> + \mathrm{O}(\ell\Delta^{\alpha-1}). \tag{117}$$

This quantity does not get small as Δ increases or ℓ decreases. In fact, most of the convection velocity will come from scales $\ell \approx L$, the integral scale. These advective effects are not scale-local, but, on the other hand, they give no net transfer of energy excitations to small scales.

4.2. Helicity cascade in 3D

The helicity cascade in 3D, first proposed by Brissaud et al. [42] and Kraichnan [43], provides also an interesting case for our analysis, with $\Pi_H = 2\Pi(\boldsymbol{\omega}, \mathbf{u}, \mathbf{u})$. Previously, we have considered only fields with Hölder exponents

strictly between 0 and 1. However, if $\mathbf{u}$ has the exponent $0 < \alpha < 1$, then $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ has exponent $-1 < \alpha - 1 < 0$! This leads to two difficulties in our previous analysis. First, the bounds that we have established may not be valid. Second, even if they hold, the condition for ultraviolet locality of the large-scale vorticity-gradient appears to be violated.

The first problem is easily addressed. In fact, the large-scale vorticity can be directly expressed in terms of velocity increments by:

$$\bar{\boldsymbol{\omega}}(\mathbf{x}) = \frac{1}{\ell} \int \mathrm{d}\mathbf{r}\ (\nabla G)_\ell(\mathbf{r}) \times [\mathbf{u}(\mathbf{x}+\mathbf{r}) - \mathbf{u}(\mathbf{x})], \tag{118}$$

Using this formula, all the estimates previously derived for Hölder continuous fields can be carried over to the vorticity field, which exists only as a distribution for $\nu \to 0$. For example,

$$\bar{\nabla}\boldsymbol{\omega}(\mathbf{x}) = -\frac{1}{\ell^2} \int \mathrm{d}\mathbf{r}\ (\nabla\nabla G)_\ell(\mathbf{r}) \times [\mathbf{u}(\mathbf{x}+\mathbf{r}) - \mathbf{u}(\mathbf{x})], \tag{119}$$

and thus

$$\bar{\nabla}\boldsymbol{\omega} = \mathrm{O}(\ell^{\alpha-2}) \tag{120}$$

when the velocity $\mathbf{u}$ has exponent α (whether pointwise or space-mean). This is the analogue of our previous estimate (35). Note that there is a more stringent requirement on the filter function for (120) to hold, namely, $\int \mathrm{d}\mathbf{r}\, |(\nabla\nabla G)(\mathbf{r})| \cdot |\mathbf{r}|^\alpha < \infty$, which involves an extra gradient. All of the estimates previously derived for Hölder continuous fields can in like manner be derived for the vorticity (or for any other gradient of a Hölder continuous field).

The second problem is also easy to address. Indeed, ultraviolet locality of the large-scale gradient should apply automatically, without any continuity condition whatsoever! In line with this reasonable expectation, the general locality estimates (45), (89) can be improved, requiring only somewhat stronger smoothness properties of the filter function G. Commuting the original filter G and the test filter Γ, we can rewrite (42) as

$$\bar{\nabla}\mathbf{u}^<(\mathbf{x}) = -\int \mathrm{d}\boldsymbol{\rho}\ \Gamma_\delta(\boldsymbol{\rho})[\nabla\bar{\mathbf{u}}(\mathbf{x}+\boldsymbol{\rho}) - \nabla\bar{\mathbf{u}}(\mathbf{x})] \tag{121}$$

As in the proof of infrared locality of the stress in Section 3.1.2, we can use the fundamental theorem of calculus to introduce an additional gradient into (121):

$$\bar{\nabla}\mathbf{u}^<(\mathbf{x}) = -\int \mathrm{d}\boldsymbol{\rho} \int_0^1 \mathrm{d}\theta\ \Gamma_\delta(\boldsymbol{\rho})\, \boldsymbol{\rho} \cdot \nabla\nabla\bar{\mathbf{u}}(\mathbf{x}+\theta\boldsymbol{\rho}). \tag{122}$$

However, in the same way as (35) [or (120)] was proved, it is also easy to show that

$$|\nabla\nabla\bar{\mathbf{u}}| = \mathrm{O}(\ell^{\alpha-2}) \tag{123}$$

whenever $\int \mathrm{d}\mathbf{r}\, |\nabla\nabla G(\mathbf{r})| < \infty$. Substituting this result into (122), we obtain

$$\bar{\nabla}\mathbf{u}^< = \mathrm{O}(\delta\ell^{\alpha-2}) \tag{124}$$

rather than (44). If we assume as before that $\bar{\nabla}\mathbf{u} \sim (\text{const.})\ell^{\alpha-1}$, we obtain

$$\frac{|\bar{\nabla}\mathbf{u}^<|}{|\bar{\nabla}\mathbf{u}|} = \mathrm{O}\left(\frac{\delta}{\ell}\right) = \frac{|\bar{\nabla}\mathbf{u}^> - \bar{\nabla}\mathbf{u}|}{|\bar{\nabla}\mathbf{u}|}. \tag{125}$$

This is another statement of ultraviolet locality of the large-scale gradient, stronger than (45) or (89). To apply this estimate to the vorticity field in the expression for the helicity flux we must assume still more smoothness of the filter function, namely, $\int \mathrm{d}\mathbf{r}\, |\nabla\nabla\nabla G(\mathbf{r})| < \infty$. General bounds even stronger than (125) can be obtained by assuming

further smoothness of the filter functions $G(\mathbf{x})$, $\Gamma(\mathbf{x})$, or, equivalently, more rapid decay of their Fourier transforms. For example, if the transforms $\hat{G}(\mathbf{k})$, $\hat{\Gamma}(\mathbf{k})$ are compactly supported in $\mathbf{k}$-space, then $\bar{\nabla}\mathbf{u}^{<} \equiv 0$ identically whenever $\delta < c\,\ell$ for some constant c.

These remarks all together imply that the helicity cascade in 3D is also local, both in the infrared and in the ultraviolet. Estimates exactly like (114) and (115) can be written down for $\Pi_H = 2\Pi(\boldsymbol{\omega}, \mathbf{u}, \mathbf{u})$.

4.3. Enstrophy cascade in 2D

The enstrophy cascade in 2D proposed by Batchelor [44] and Kraichnan [34] turns out to be the most subtle of all the cases that we shall examine in this paper. In this example, $\Pi_Z = \Pi(\omega, \omega, \mathbf{u})$, where $\mathbf{u}$ is the fluid velocity and ω is the pseudoscalar vorticity in 2D. However, recall that Batchelor and Kraichnan predicted for this cascade a k^{-3} energy spectrum (with later a logarithmic correction [41]). This spectrum corresponds to a velocity scaling exponent $\alpha_2 = 1$ or $\mathbf{u} \in B^1_2$ and, likewise, $\omega \in B^0_2$. For example, see [45]. The zero-index Besov space consists of distributions, not continuous (or even square-integrable) functions. For example, the pure k^{-3} spectrum leads to a mean-square vorticity-increment which scales as

$$\langle[\delta\omega(\mathbf{r})]^2\rangle \sim (\text{const.})\eta^{2/3} \ln\left(\frac{r}{\ell_d}\right), \tag{126}$$

for $\ell_f \gg r \gg \ell_d$, where ℓ_f is the forcing length-scale, η is the mean enstrophy flux and $\ell_d = \nu^{1/2}\eta^{-1/6}$ is the Kraichnan dissipation length. But (126) does not even exist as viscosity $\nu \to 0$ and thus $\ell_d \to 0$. The two-point correlation for $\ell_f \gg r \gg \ell_d$ (with γ_E Euler's constant)

$$\langle\omega(\mathbf{r})\omega(0)\rangle \sim (\text{const.})\eta^{2/3}\left[\ln 2 - \gamma_E - \ln\left(\frac{r}{\ell_f}\right)\right], \tag{127}$$

does exist as $\nu \to 0$, when $r > 0$, but the mean enstrophy $\langle\omega^2\rangle \sim (\text{const.})\eta^{2/3}\ln(\ell_f/\ell_d)$ diverges and this is responsible for the divergence in (126). In general, all of the pth-order correlation functions $\langle\omega(\mathbf{r}_1)\cdots\omega(\mathbf{r}_p)\rangle$ are expected to exist as $\nu \to 0$ when the points are distinct, $\mathbf{r}_i \neq \mathbf{r}_j$ for $i \neq j$, but to exhibit logarithmic singularities when $|\mathbf{r}_i - \mathbf{r}_j| \to 0$ for some $i \neq j$ [46]. Thus, it is expected that $\omega \in B^0_p$ and $\mathbf{u} \in B^1_p$ for general $p \geq 2$.

This situation creates several difficulties. First, the same problems that occur for the 3D helicity cascade occur here as well. However, these problems are not serious and can be handled exactly as for the vorticity in the helicity cascade. Namely, (118) can be used to derive estimates for $\bar{\omega}$ and ultraviolet locality of $\bar{\nabla}\omega$ requires in fact no Hölder continuity of ω. On the other hand, new problems arise for $\boldsymbol{\tau}(\omega, \mathbf{u})$, because the exponent 0 of the vorticity is exactly on the borderline for ultraviolet locality and the exponent 1 of the velocity is exactly on the borderline for infrared locality.

On the face of it, the more serious of these problems is the 0 exponent and possible ultraviolet nonlocal effect of the vorticity field. A priori it is not even clear that $\boldsymbol{\tau}(\omega, \mathbf{u})$ exists for $\nu \to 0$, because of the asymptotic behavior in (126). It appears that there may be a serious ultraviolet divergence. This is the manifestation of the fact that a vorticity field $\omega \in B^0_2$ exists only as a distribution. Thus, $\boldsymbol{\tau}(\omega, \mathbf{u})$ can only exist as $\nu \to 0$ if there are significant cancellations in the space-average with respect to the filter function G_ℓ. The possibility of such cancellations was already recognized by Kraichnan [41], who wrote of the spectral enstrophy flux $\Pi_Z(k)$ that "the effects of wavenumbers $\gg k$ can plausibly be expected to average out over scales $\sim 1/k$ and over times the order of the characteristic distortion time ω_k^{-1}." In fact, numerical simulations [47] have indicated that $\boldsymbol{\tau}(\omega, \mathbf{u})$ is very strongly ultraviolet *local*. This is beyond what we can prove by our simple estimates, which do not take into account the required cancellations. It is important to keep in mind that our conditions for scale-locality are sufficient, but not necessary.

It turns out that the really serious problem is the unit exponent 1 and associated infrared nonlocal effect of the velocity field $\mathbf{u}$ in $\boldsymbol{\tau}(\omega, \mathbf{u})$. Recall that the main estimate used in Section 3.1.2 to establish infrared locality of $\boldsymbol{\tau}(\omega, \mathbf{u})$

was (58) or $|\delta\mathbf{u}^{>}(\mathbf{r})| = \mathrm{O}(r\Delta^{\alpha-1})$. In the present case this corresponds to

$$|\delta\mathbf{u}^{>}(\mathbf{r})| = \mathrm{O}(r \ln^{p} \Delta), \tag{128}$$

for some $p > 0$, since $\alpha = 1$. However, when (126) holds, then

$$|\delta\mathbf{u}(\mathbf{r})| \sim (\text{const.})\eta^{1/3} r \ln^{p} r, \tag{129}$$

and this is (at most) only logarithmically larger than (128). Thus, the contributions of the large-scale velocity increments are non-negligible and contribute significantly to enstrophy flux. In this case there is no possibility of cancellations, because the large-scale contribution is coherent across the small-scale region of diameter $\sim\ell$. Of course, this view of the enstrophy cascade as highly infrared nonlocal agrees with the conventional wisdom [34,41].

5. Relations to previous work

It is interesting to compare our analysis with some of the prior work on the locality issue.

5.1. *In the beginning: Kolmogorov and Onsager*

Kolmogorov in his first 1941 paper [30] made a few remarks on scale-locality, in an extended footnote. He stated there that energy cascade should be local in the sense that it should involve transfer "successively" from "pulsations" of one order to those of the next higher order. The mechanism of the energy transfer between "pulsations" that Kolmogorov invoked is not so clear, but involved some unstable breakdown of the large structures into small ones. His brief remarks do contain one very concrete statement that "the differences $w_\alpha(P) = u_\alpha(P) - u_\alpha(P^{(0)})$ of the velocity components in neighboring points P and $P^{(0)}$ of the four-dimensional space (x_1, x_2, x_3, t) are determined nearly exclusively by pulsations of higher orders." [30]. This quotation shows that Kolmogorov realized the infrared locality of velocity increments, in our sense.

Onsager, in another foundational paper on turbulence theory [1], had a bit more to say on the subject:

"Experience indicates that for large REYNOLDS numbers the over-all rate of dissipation is completely determined by the intensity $\overline{v^2}$ together with the 'macroscale' L of the motion, and that the viscosity plays no primary role except through the condition that the REYNOLDS number

$$\mathcal{R} = (\overline{v^2})^{1/2} \frac{\rho L}{\eta} \tag{10}$$

must be sufficiently large. Under the circumstances, dimensional considerations uniquely determine the law of dissipation

$$Q = -\frac{\mathrm{d}}{\mathrm{d}t}\overline{v^2} = (\text{const.})\frac{(\overline{v^2})^{3/2}}{L}, \tag{11}$$

and this has been verified by many experiments ....

In order to understand the law of dissipation described by (11), which does not involve the viscosity at all, we have to visualize the redistribution of energy as an accelerated cascade process. If we write the right member of (11) in the form of a product:

$$\overline{v^2}\left(\frac{\overline{v^2}}{L^2}\right)^{1/2};$$

then the first factor represents the energy density, and most of this belongs to wave-numbers of the order $1/L$. The second factor is a rate of shear—not the overall rate of deformation of the fluid, but only that part of it which belongs to motion on the largest scale. Indeed, it is not difficult to see that the modification of a smooth current by a fine-grained disturbance will depend on the total displacements involved, rather than on the rate of shear.

...Similar reasoning may be applied to subsequent steps in the redistribution process, and we are led to expect a cascade such that the wave-numbers increase typically in a geometric series, by a factor of the order 2 per step. The energy is reprocessed through a given range of wave-numbers mainly with the aid of velocity gradients which belong to wave-numbers of the same order of magnitude." [1]

This passage, despite some obscure parts, has many points of contact with our analysis. Foremost, Onsager's two factors in the large-scale energy flux, $\overline{v^2}$ and $(\overline{v^2}/L^2)^{1/2}$, are in exact correspondence to our two terms τ_{ij} and $\bar{S}_{ij}$, respectively. Of course, this "flux-force" form of dissipation was quite familiar to Onsager from his pioneering work on non-equilibrium thermodynamics [48]. We see also a clear statement that the energy density and velocity gradients responsible for the transfer are scale-local. Furthermore, the latter depends upon "displacements" or increments rather than exact derivatives, in agreement with our analysis. Some of the obscure points in the above passage are made more clear in a version of the same argument in a letter of Onsager to C.C. Lin four years earlier [49]. We shall not quote that letter at length (see [50] for the full text) but just note that Onsager stated there a condition

$$\overline{v^2} < \infty; \ \overline{(\nabla \times v)^2} = \infty$$

and then noted that, "with a hypothesis slightly stronger ... the motion which belongs to wave-numbers of the same order of magnitude as k itself will furnish the greater part of the effective rate of shear." Thus, Onsager recognized that some regularity of the velocity is required for locality, similar to our condition that the Hölder exponent α lie between 0 and 1.

5.2. Kraichnan and analytical closure

As far as we are aware, the first quantitative results on scale-locality were obtained by Kraichnan in 1959, discussing energy conservation within his DIA closure. He wrote:

"The formal property [of conservation] has meaning, however, only if the integrals involved converge properly, which corresponds physically to the presumed localness of cascade. We may verify the latter property by considering the total power input to all modes of wavenumber $k'' > k'$ from direct interactions with all mode pairs p, q such that p or $q < k \ll k'$, where k and k' are fixed wave-numbers and all wave-numbers concerned are within the inertial range. Using $r(k, s) = g(k, |s|) = J_1(2v_0ks)/(v_0ks)$ [the DIA solution] and computing this power as

$$\Pi(k'|k) = \frac{1}{2}\int_{k'}^{\infty} \mathrm{d}k'' \int \int_{\Delta''}^{\{p \text{ or } q<k\}} S(k''|p, q)\,\mathrm{d}p\,\mathrm{d}q \tag{6.4}$$

we find, after considerable algebra, the asymptotic result

$$\Pi(k'|k) = (\text{numerical factor})[f(0)]^2\epsilon\left(\frac{k}{k'}\right)^{3/2} \quad (k \ll k'). \tag{6.5}$$

The triad interactions involved in the integration are shown in Fig. 4. $\Pi(k'|k)$ goes to zero with k/k', so that the energy transport is asymptotically local, as originally assumed. However, the dependence on k/k' is not particularly strong, and thus the cascade is rather diffuse." [51]

Kraichnan's estimate in his Eq. (6.5) expresses the property of *infrared locality* of energy cascade.

In general, analytical closures of the DIA class (e.g. TFM, EDQNM, etc.) give a result of the form

$$\Pi(k'|k) \sim (\text{const.})\epsilon \left(\frac{k}{k'}\right)^{3-n} \qquad (k \ll k') \tag{130}$$

when the energy spectrum $E(k) \sim (\text{const.})k^{-n}$. See [40,52]. Thus, the condition for infrared locality is $n < 3$. Since the spectral exponent n and the Hölder exponent α are related by $n = 1 + 2\alpha$ for a monofractal field, we see that this corresponds to our condition $\alpha < 1$. One can likewise define the quantity $\Pi(k'|k)$ for $k > k'$ by using instead the condition $\{p \text{ or } q > k\}$ in Kraichnan's Eq. (6.4) above. This quantity then measures *ultraviolet locality* of the energy cascade. Analytical closures give [40,52]:

$$\Pi(k'|k) \sim (\text{const.})\epsilon \left(\frac{k'}{k}\right)^{2(n-1)} \qquad (k \gg k'). \tag{131}$$

The reason for the factor of 2 in the exponent is that, by wavenumber conservation, $p > k \gg k'$ implies also $q \approx p \gg k'$ and vice versa, so there must always be two large wavenumbers in the triad simultaneously. Thus, the condition in analytical closures for ultraviolet locality is $n > 1$, corresponding to our condition $\alpha > 0$. Notice that the closure estimates of nonlocal contributions in (130) and (131) go to zero faster than our corresponding estimates (114) and (115), namely, as the *square* of our bounds. This makes good sense since our estimates are for individual realizations with no statistical averaging, whereas the latter leads to additional cancellations in the incoherent contributions of nonlocal triads to energy flux.

A more recent work of L'vov and Falkovich [53] has attempted the ambitious task of obtaining asymptotic estimates on nonlocal contributions to mean transfer, by direct statistical averaging over Navier–Stokes solutions without the ad hoc assumptions of the analytical closures [40,52]. The approach of [53] is less rigorous than ours but complementary in certain respects. It is weaker in the sense that it resorts to statistical averaging but stronger in the sense that it investigates as well the scale-locality of (quasi-Lagrangian) time-scales. In the present study we have established locality of instantaneous transfer for individual flow realizations, without consideration of multi-time statistics.

In general, the results of the investigation in [53] are consistent with ours, except in one important respect. Those authors claimed to obtain "counterbalanced locality", in the sense that we discussed in Section 4.1, for any scaling exponents and not just for the K41 mean-field value. More precisely, they claimed for IR nonlocal transfer across wavenumber k a scaling law $(k\Delta)^{\zeta_2-2}$ when $k\Delta \gg 1$ and for UV nonlocal transfer a similar scaling law $(k\delta)^{2-\zeta_2}$ when $k\delta \ll 1$, in our notations. This differs from the result for analytical closures, embodied in our Eqs. (130) and (131), which are "counterbalanced" if and only if $3 - n = 2(n - 1)$ or $n = 5/3$. Furthermore, IR locality in the analytical closures requires $n < 3$ and UV locality requires $n > 1$. However, it was claimed in [53] that, while IR locality requires $\zeta_2 < 2$ or Hölder exponent $h < 1$, UV locality "gives no restrictions for the possible values of exponent h." The authors of [53] recognized these discrepancies with the closure results and speculated that the "difference between the initial result of Kraichnan [7] and the present approach is due to the fact that the direct-interaction approximation is an uncontrolled procedure."

However, the results in [53] conflict not only with DIA but also with our rigorous estimates. In particular, we obtained a requirement on velocity Hölder exponent α – whether pointwise or space-mean – that it be positive ($\alpha > 0$) for UV locality to hold. Although we derive only upper bounds and not asymptotic scaling results, our estimate (115) contradicts the scaling law of [53] for UV nonlocal transfer whenever $2\zeta_3 + 3\zeta_2 > 6$, or, in a monofractal case, whenever $\alpha > 1/2$. The more interesting question, of course, is whether UV locality is actually violated if $\alpha < 0$, i.e. if negative Hölder exponents develop. We cannot conclude this on the basis of our estimates, which are upper bounds only. However, we can conclude that the scaling of UV nonlocal transfer in [53] is erroneous in general. Without attempting a detailed diagnosis of their very intricate diagrammatic analysis, we note only that order-by-order results in a non-convergent perturbation expansion cannot be considered to be "exact."

It is important that we have established locality for individual flow realizations without statistical averaging, since, otherwise, there would be little basis to the idea of universal statistics at small scales. Furthermore, we have made no closure hypotheses of approximate validity, but simply employed regularity assumptions that are implied by experiment. On the other hand, our estimates, like those from closures, show that the cascade is, as Kraichnan expressed it, only "asymptotically local" and "diffuse". The rate of vanishing of the nonlocal contributions with increasing scale-separation are quite slow power-law decays. Thus, very long inertial ranges might be required for local transfer to dominate.

5.3. Direct numerical simulation and previous exact results

Despite the results presented in this paper, a number of previous DNS studies have found that 3D energy cascade, as measured by the spectral flux $\Pi_E(k)$, is dominated by "local transfer through nonlocal triads" [54–56]. These interactions give a contribution to flux which is proportional to the large-scale advecting velocity and not to the large-scale strain. How can these findings be consistent with our claims of scale-locality? In fact, the sharp spectral filter $\hat{G}(\mathbf{k}) = \theta(1 - |\mathbf{k}|)$ [where θ is the Heaviside step function] does not satisfy the conditions for our estimates, since it decays only very slowly $\sim|\mathbf{r}|^{-1}$ in physical space and also takes on negative values. As shown by explicit examples in [5], it is not just that the proofs break down but that the estimates themselves are invalid. This confirms the numerical results found in [54–56].

However, these results are due simply to the use of $\Pi_E(k)$, which is an inappropriate measure of energy transfer. The phenomenon was already discussed by Kraichnan in his LHDIA closure, where he wrote that

"Convection of high-wavenumber structures by strongly excited low-wavenumber velocity components implies a rapid change of phase of the high-wavenumber Fourier amplitudes; that is to say, a rapid exchange of energy between sine and cosine components of the high wavenumbers. This exchange is represented in (2.6) [the LHDIA energy balance] by the large cancelling input and output contributions. The net contribution $T_{<q}(k, t)$ represents the effect of straining alone." [57]

The same conclusion was reached in [5] without the use of any closure approximation. It was pointed out there that the spectral flux $\Pi_E(k)$ is not sensitive to the *distance* in k-space that energy is transferred by given classes of triads and hence does not really measure their efficiency for spectral transport. Local triads move the energy typically from k to $2k$ (these are like "giant steps" in the familiar children's game of "Mother-May-I?"). However, "local transfer through nonlocal triads" as discussed in [54–56] moves energy only from k to $k + q$, where q is a small wavenumber characteristic of the energetic large scales. These are like tiny "baby steps" and, furthermore, they have no coordinated direction and tend to cancel out. It was found in [5] that in a scale-average of the spectral flux over an octave band

$$\bar{\Pi}_E^{(N)} = \frac{1}{2^N} \int_{2^N}^{2^{N+1}} \mathrm{d}k\, \Pi_E(k) \tag{132}$$

these many small transfers nearly cancel out to give a net contribution proportional to the strain in the large scales. Similar arguments were made also in [58,59] that nonlocal contributions should largely cancel when summed over classes of wavenumber triads.

The filtering approach automatically incorporates such cancellations. We have already seen that, in that formalism, the energy flux in physical space is proportional to the large-scale strain and is locally determined in scale. Further insight comes from writing its space-average in terms of Fourier modes. An easy computation gives

$$\int \mathrm{d}\mathbf{x}\, \Pi_E(\mathbf{x}) = \int_0^\infty \mathrm{d}k\, P_\ell(k)\, \Pi_E(k) \tag{133}$$

with

$$P_\ell(k) = -\frac{\mathrm{d}}{\mathrm{d}k}|\hat{G}(k\ell)|^2.$$

It follows that

$$P_\ell(k) \geq 0, \int_0^\infty \mathrm{d}k\, P_\ell(k) = 1 \tag{134}$$

for any spherically-symmetric filter G such that $|\hat{G}(k)|$ is monotonically decreasing in k. Thus, the global flux in the filtering approach is just the spectral flux $\Pi_E(k)$ scale-averaged around the wavenumber $\sim 1/\ell$ with an averaging distribution of width $\sim 1/\ell$ in k-space. Hence, (133) for reasonable filters is essentially identical to (132) and shows the same cancellations.

6. Concluding remarks

The results we have established here have several important implications and applications. The ultraviolet locality of stress is fundamental for LES of turbulent fluids, since it implies that only adjacent subgrid modes contribute dominantly to the stress. This fact can be made the basis of a convergent approximation to the stress, which provides a turbulent constitutive relation for LES. This application has been developed in detail elsewhere [60]. We should note that other forms of locality may be important for turbulence dynamics, such as the scale-locality of time-scales or of "eddy-turnover times." To prove this would require a more difficult analysis than the essentially kinematic one presented here. The paper [53] identifies many of the key issues involved.

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